

TICVAI

Resource Management Configuration

Backend Structure Module

PROFESSIONALLY FORMATTED EDITION

Comprehensive functional specification covering resource configuration, availability, workforce, rostering, operations, assets, events, AI optimization, mobile operations, governance, and analytics.

Document type

Functional & Backend Configuration Reference

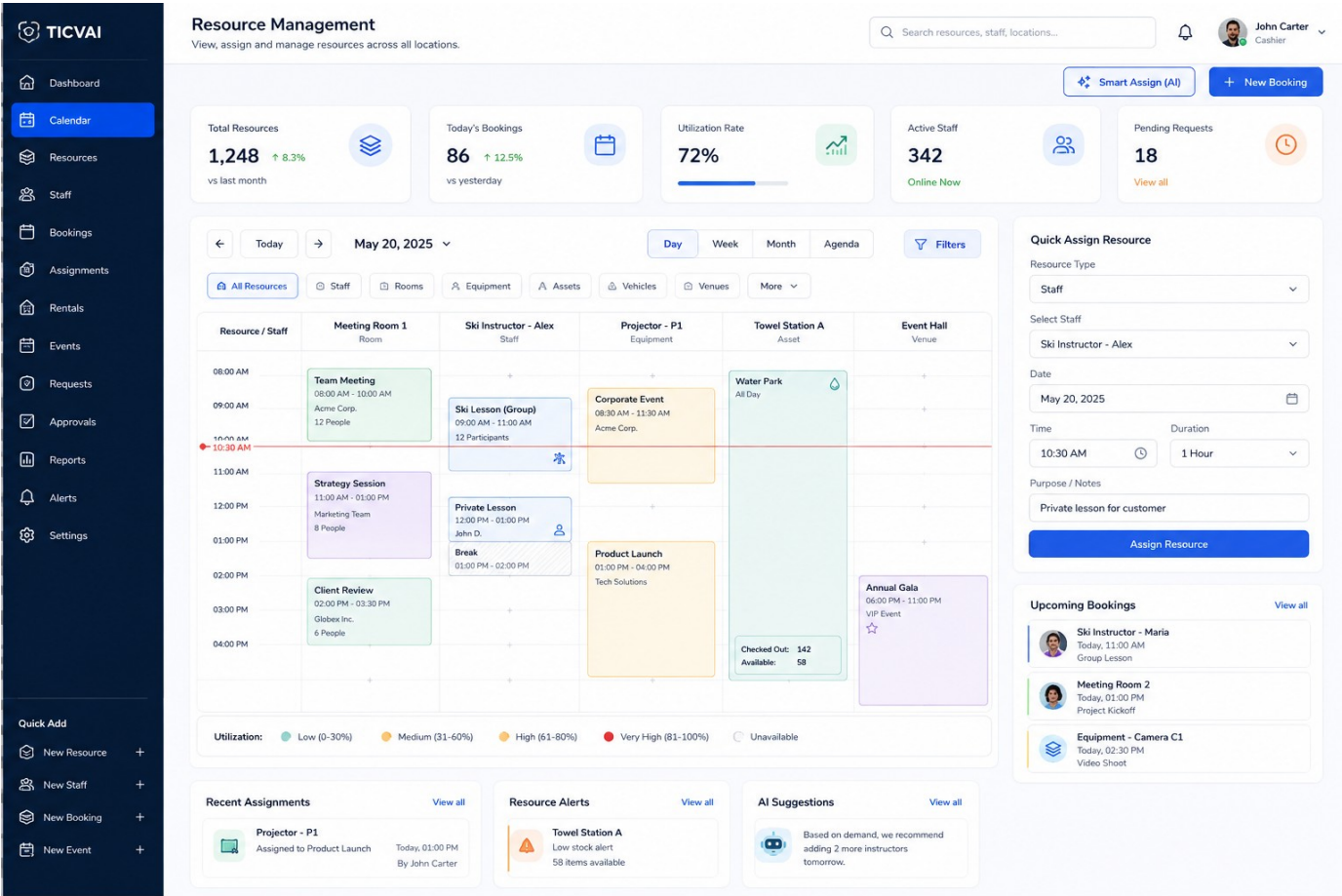
Module

Resource Management

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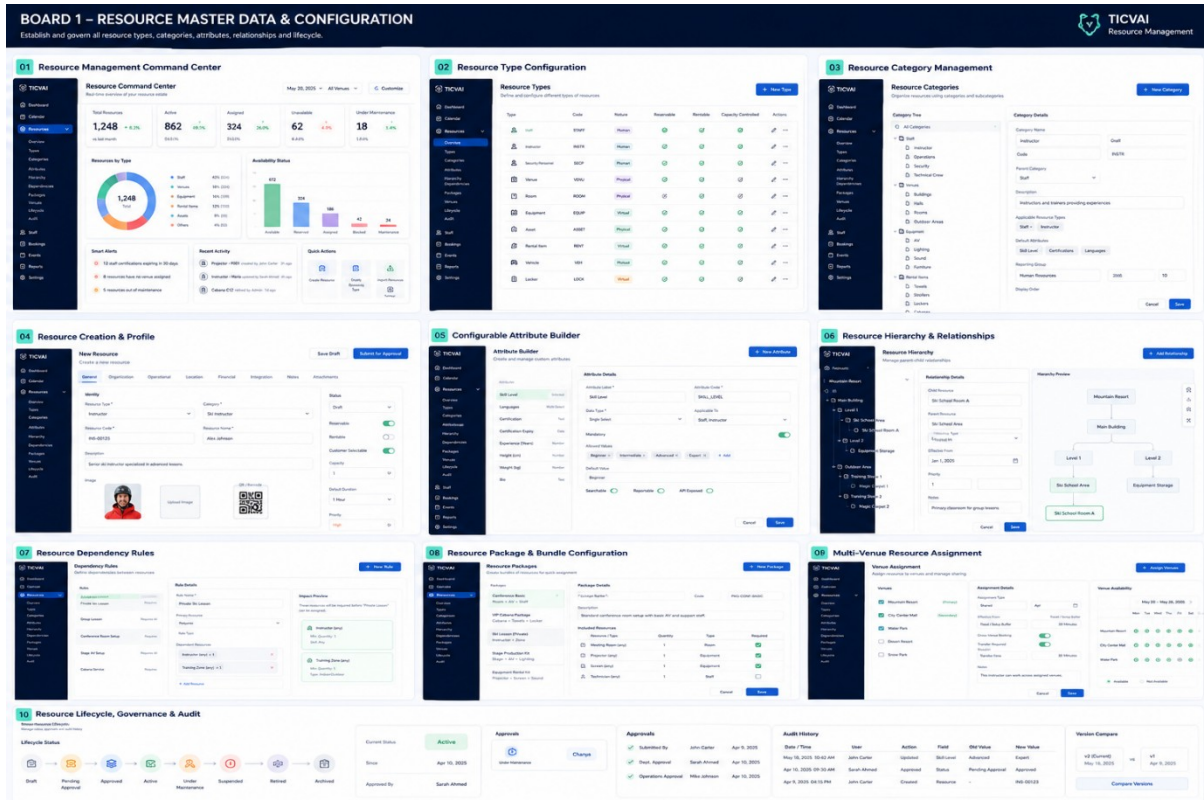
TICVAI Resource Management



Board 1 — Resource Master Data & Configuration

Board Objective

Provide administrators with a centralized, intelligent backend workspace to create, classify, configure, organize, govern, and maintain every resource that can be used anywhere within the TICVAI platform.



A resource may represent:

- Staff or personnel
- Venue
- Hall
- Room
- Stage
- Area
- Attraction space
- Equipment
- Asset
- Rental item
- Vehicle
- Locker
- Cabana
- Meeting room
- Stroller
- Towel
- AV equipment
- Sound or lighting equipment
- Any customer-defined resource type

The board must support resource reuse across ticketing, experiences, events, rentals, workforce scheduling, POS, mobile applications, and APIs.

Primary matrix coverage: 1.2.1, 1.2.2, 1.2.23–1.2.26, 1.2.44, 1.2.46, 1.2.72, 1.2.75, 1.2.77, 1.2.78.

Screen 01 — Resource Management Command Center

Purpose

Provide administrators and operational managers with the main entry point for Resource Management and an instant overview of the organization's complete resource inventory.

Backend Configuration

The screen shall display configurable KPI widgets including:

- Total active resources
- Resources by type
- Resources by venue
- Available resources
- Assigned resources
- Reserved resources
- Resources under maintenance
- Suspended resources
- Retired resources
- Resources with expiring certifications
- Resources with unresolved configuration issues
- Recently created resources
- Recently modified resources

Administrators shall be able to filter the command center by:

- Tenant
- Business unit
- Country
- Venue
- Resource type
- Category
- Status
- Owner
- Department
- Tags
- Availability status
- Effective date

The screen shall provide quick actions for:

- Create resource
- Create resource type
- Create category
- Import resources
- Duplicate resource
- Assign resource to venue
- Create package
- Review configuration warnings

An AI summary panel should highlight conditions such as:

- Unused resources
- Duplicate resources
- Missing attributes
- Resources approaching retirement
- Resources without venue assignments
- Unresolved resource dependencies

Business Rules

Dashboard information must respect the logged-in user's tenant, venue, department, and permission scope.

Users must only see resources they are authorized to manage.

Acceptance Condition

Authorized users can view the complete permitted resource estate, filter and search resources, identify configuration issues, and launch governed resource-management actions directly from the command center.

Screen 02 — Resource Type Configuration

Purpose

Allow administrators to define the different classes of resources supported by the organization without requiring software development.

Backend Configuration

Administrators shall be able to create configurable resource types such as:

- Staff
- Instructor
- Security personnel
- Venue
- Room
- Hall
- Area
- Cabana
- Equipment
- Asset
- Rental item
- Vehicle
- Locker

Each resource type shall support:

- Resource type code
- Resource type name
- Description
- Icon
- Display color
- Resource nature
- Physical / human / virtual classification
- Reservable yes/no

- Rentable yes/no
- Capacity-controlled yes/no
- Schedule-controlled yes/no
- Inventory-controlled yes/no
- Staff qualification required yes/no
- Maintenance-controlled yes/no
- Check-in/check-out supported yes/no
- Deposit applicable yes/no
- Customer selectable yes/no
- API visible yes/no

Administrators shall determine which configuration sections are applicable to each resource type.

Example:

A **Staff** resource may require skills, certifications, shifts, breaks, and attendance.

A **Cabana** resource may require capacity, location, opening hours, price linkage, and availability.

A **Towel** resource may require inventory, checkout, deposit, return, and condition tracking.

Governance

Changes to resource-type definitions may require approval depending on tenant policy.

Existing resources must be evaluated when their resource type is changed.

Acceptance Condition

Administrators can create reusable resource types and define which operational capabilities and attributes apply to each type without code changes.

Screen 03 — Resource Category Management

Purpose

Provide a flexible classification structure beneath resource types so resources can be organized, searched, reported, and governed consistently.

Backend Configuration

Administrators shall create categories and subcategories such as:

- **Equipment**
 - AV
 - Lighting
 - Sound
 - Furniture
- **Staff**
 - Instructor
 - Operations
 - Security
 - Technical Crew
- **Rental**
 - Towels
 - Strollers
 - Lockers

- Cabanas

The screen shall support:

- Category code
- Category name
- Description
- Parent category
- Applicable resource types
- Display order
- Tags
- Reporting group
- Cost center
- Default attributes
- Default availability policies
- Default approval workflow
- Active/inactive status

Categories may inherit configuration from parent categories.

Administrators should be warned when changing a category used by active resources.

Acceptance Condition

Resources can be grouped into configurable hierarchical categories and subcategories that can subsequently be used throughout search, scheduling, reporting, automation, and integrations.

Screen 04 — Resource Creation & Profile

Purpose

Provide the principal workspace for creating and maintaining an individual resource.

Backend Configuration

The resource profile shall contain configurable sections including:

Identity

- Resource ID
- Resource code
- Resource name
- Short name
- Resource type
- Category
- Description
- Image
- Barcode / QR identifier
- External reference ID

Organization

- Tenant
- Business unit
- Venue

- Department
- Owner
- Responsible manager
- Cost center

Operational Configuration

- Active status
- Reservable status
- Rentable status
- Capacity
- Unit of measure
- Customer selectable
- Priority
- Availability mode
- Scheduling mode
- Default duration
- Minimum booking duration
- Maximum booking duration

Location

- Venue
- Building
- Floor
- Zone
- Area
- GPS/location metadata where applicable

Commercial & Financial

- Internal cost
- Hourly cost
- Daily cost
- Replacement value
- Deposit requirement
- Tax treatment
- Accounting reference

Integration

- External system ID
- API identifier
- Integration source
- Last synchronization status

The resource profile shall support:

- Save as draft
- Submit for approval

- Activate
- Suspend
- Clone
- Archive
- Retire

AI Assistance

AI should be able to suggest:

- Resource category
- Required attributes
- Tags
- Similar existing resources
- Potential duplicates

Acceptance Condition

Authorized users can create and maintain a complete resource profile, save it as draft, validate its configuration, submit it through governance, and activate it for downstream use.

Screen 05 — Configurable Attribute Builder

Purpose

Allow customers to extend resource records with their own attributes without changing the core application.

Backend Configuration

Administrators shall create configurable attributes such as:

- Skill level
- Height
- Weight
- Seating capacity
- Equipment model
- Serial number
- Size
- Manufacturer
- Color
- Power requirement
- Warranty expiry
- Maximum occupancy
- Language
- Certification level
- Rental condition

Supported data types shall include:

- Text
- Number
- Decimal
- Currency
- Date

- Date/time
- Boolean
- Single select
- Multi-select
- Lookup
- Attachment
- URL
- Measurement
- Formula/calculated field

For each attribute, administrators shall configure:

- Attribute code
- Attribute label
- Applicable resource type
- Applicable category
- Mandatory/optional
- Default value
- Allowed values
- Minimum/maximum
- Validation expression
- Searchable yes/no
- Reportable yes/no
- API exposed yes/no
- Customer visible yes/no
- Effective date
- Display sequence

Conditional attributes shall be supported.

Example:

If **Resource Type = Vehicle**, display vehicle registration and passenger capacity.

If **Resource Type = Instructor**, display skill level, languages, and qualifications.

Acceptance Condition

Administrators can extend resource records with reusable, validated, type-specific attributes without requiring application development.

Screen 06 — Resource Hierarchy & Parent-Child Relationships

Purpose

Represent physical and operational relationships between resources.

Backend Configuration

Administrators shall be able to create parent-child relationships such as:

Venue

- Building
- Floor
- Hall
- Room

or:

AV System

- Projector
- Screen
- Speakers
- Microphones

The screen shall provide:

- Tree view
- Expand/collapse hierarchy
- Drag-and-drop organization
- Parent selection
- Child-resource assignment
- Relationship type
- Effective date
- Relationship priority
- Dependency indication

Supported relationship examples:

- Contains
- Belongs to
- Located in
- Operated by
- Supported by
- Part of
- Dedicated to

The system shall prevent:

- Circular hierarchies
- Invalid parent relationships
- Cross-tenant relationships without authorization

Operational Impact

Hierarchy may influence:

- **Availability**
- Capacity
- Maintenance
- Reporting
- Reservations
- Event allocation

Example:

If an entire hall is closed for maintenance, resources configured as dependent children may also become unavailable.

Acceptance Condition

Administrators can create and manage multi-level resource hierarchies, and the platform validates those relationships before they are used operationally.

Screen 07 — Resource Dependency Rules

Purpose

Define operational relationships where one resource requires, depends upon, conflicts with, or influences another resource.

Backend Configuration

Administrators shall be able to configure dependencies such as:

- Requires
- Requires one of
- Requires all
- Conflicts with
- Cannot operate simultaneously
- Preferred with
- Substitute for
- Backup for
- Shares capacity with

Example:

Stage A

requires:

- Sound System A
- Lighting Rig A

Private Ski Lesson

requires:

- One instructor
- One training zone

Dependency rules shall support:

- Minimum quantity
- Maximum quantity
- Mandatory/optional
- Priority
- Effective dates
- Venue limitation
- Time overlap rules
- Availability dependency
- Capacity dependency

The system must evaluate dependencies before confirming assignments or reservations.

Conflict Detection

The system shall display:

- Missing dependencies
- Resource conflicts
- Circular dependency warnings
- Unavailable dependent resources

Acceptance Condition

Administrators can define reusable dependency rules, and the system validates those dependencies before affected resources can be reserved or assigned.

Screen 08 — Resource Package & Bundle Configuration

Purpose

Allow commonly used combinations of resources to be predefined and assigned as a single operational package.

Backend Configuration

Examples may include:

- **Conference Package**
 - Meeting Room A
 - Projector
 - Screen
 - Microphone
 - Technician
- **VIP Cabana Package**
 - Cabana
 - Towels
 - Locker
 - Service attendant
- **Event Stage Package**
 - Stage
 - Sound equipment
 - Lighting
 - Technician
 - Security

Administrators shall configure:

- Package code
- Package name
- Description
- Applicable venues
- Included resources
- Included resource types
- Required quantities
- Mandatory/optional components
- Substitute resources
- Allocation priority
- Effective dates
- Minimum/maximum duration
- Approval requirement
- Internal package cost

Packages should support:

- Fixed resources

- Dynamic resources
- Skill-based staff resources
- Resource-type placeholders

Example:

Instead of assigning **John Smith**, the package may request:

1 × Technician with AV Level 2 certification.

The allocation engine can then select the most suitable available technician.

Acceptance Condition

Administrators can create reusable resource packages containing fixed or dynamically selected resources and use those packages later during experience, event, reservation, and operational allocation.

Screen 09 — Multi-Venue Resource Assignment

Purpose

Control where resources may operate and whether they can be shared across venues.

Backend Configuration

The screen shall support:

- Primary venue
- Secondary venues
- Shared resource pool
- Venue availability
- Department ownership
- Effective dates
- Travel/setup buffer
- Cross-venue booking permission
- Resource transfer requirement
- Transfer time
- Venue-specific restrictions
- Venue-specific cost

Example:

A portable projector may belong to **Venue A** but be shared with **Venue B**.

An instructor may work across several attractions or venues.

Administrators shall determine whether the resource is:

- Venue-exclusive
- Venue-shared
- Tenant-wide
- Temporary transfer
- Centrally pooled

Validation

The platform shall check for:

- Overlapping assignments
- Travel/setup time

- Venue authorization
- Venue-specific qualification requirements

Acceptance Condition

Resources can be assigned to one or multiple venues under controlled rules, with conflicts and transfer constraints validated before operational allocation.

Screen 10 — Resource Lifecycle, Governance & Audit

Purpose

Manage the entire operational lifecycle of a resource from creation through retirement while maintaining full traceability.

Lifecycle States

The platform shall support configurable states such as:

- Draft
- Pending Approval
- Approved
- Active
- Temporarily Unavailable
- Under Maintenance
- Suspended
- Retired
- Archived

Administrators shall configure:

- Allowed status transitions
- Approval requirements
- Effective dates
- Retirement reason
- Suspension reason
- Replacement resource
- Disposal information
- Depreciation reference
- Asset-retirement information

Governance

Actions may require:

- Maker-checker approval
- Manager approval
- Finance approval
- Operations approval

Permission controls shall determine who may:

- Create
- Edit
- Approve
- Activate

- Suspend
- Retire
- Archive

Audit History

The immutable audit timeline shall record:

- User
- Timestamp
- Action
- Previous value
- New value
- Approval
- Reason
- Source channel
- API/system origin
- Correlation/reference ID

Users shall be able to compare versions of a resource configuration.

Acceptance Condition

The complete lifecycle of every resource is governed by permissions and configurable workflows, while every material configuration change remains permanently traceable through immutable audit history.

Board 1 Shared Rules

All ten screens shall support, where applicable:

- Multi-tenant architecture
- Multi-venue configuration
- Role-based and policy-based access control
- Arabic and English localization
- Configurable approval workflows
- Effective dating
- Version history
- API accessibility
- Event-driven integration
- Search and filtering
- Import/export controls
- Audit history
- AI-assisted configuration
- Responsive administration interface

No resource shall become operationally available until all mandatory configuration and validation requirements have been satisfied.

Board 1 End-to-End Outcome

After completing Board 1 configuration, TICVAI shall have a standardized **Resource Digital Twin** for every resource in the organization.

Each resource will have:

Identity → Type → Category → Attributes → Hierarchy → Dependencies → Package Membership → Venue Assignment → Lifecycle → Audit

This resource master then becomes the foundation used by:

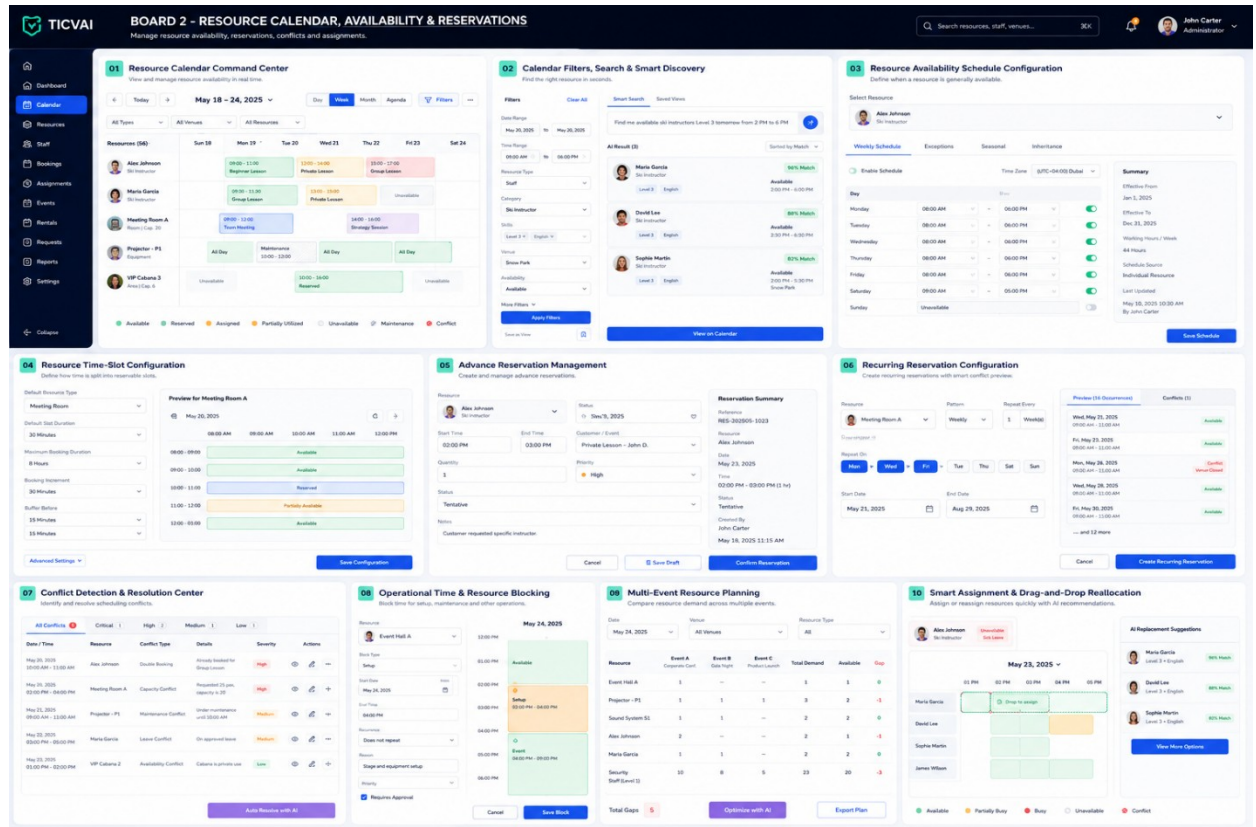
- Calendar & availability
- Staff scheduling
- Rostering
- Experience assignment
- Ticketing
- Rentals
- Events
- AI allocation
- Mobile workforce
- Analytics
- APIs

TICVAI Resource Management

Board 2 — Resource Calendar, Availability & Reservations

Board Objective

Provide TICVAI with a centralized, real-time **Resource Scheduling & Availability Engine** where authorized users can visually see when resources are available, reserved, assigned, unavailable, under maintenance, or in conflict and can allocate or reassign them directly from the calendar.



The calendar must operate as an **interactive operational workspace**, not simply a calendar display.

Users should be able to:

- View resources across time
- Search and filter resources
- See real-time availability
- Understand utilization visually
- Select an available resource directly from the calendar
- Create reservations
- Make advance reservations
- Configure recurring reservations
- Detect scheduling conflicts
- Block operational/setup/maintenance periods
- Manage simultaneous events
- Drag and drop assignments
- Automatically find replacement resources

- Receive AI recommendations

Primary matrix coverage: 1.2.6–1.2.7, 1.2.13, 1.2.18–1.2.22, 1.2.27–1.2.31.

Related coverage: 1.2.12, 1.2.19, 1.2.29, 1.2.52–1.2.57, 1.2.72–1.2.73, 1.2.77.

Screen 01 — Resource Calendar Command Center

Purpose

Provide the main operational calendar from which users can see and manage resource availability, reservations, assignments and operational blocks.

Calendar Views

The interface shall support:

- Timeline
- Day
- Week
- Month
- Agenda
- Resource view

Resources may appear as rows or columns depending on the selected calendar mode.

Users shall be able to switch between:

- All Resources
- Staff
- Rooms
- Areas
- Venues
- Equipment
- Assets
- Rental Resources
- Vehicles
- Custom Resource Types

Visual Resource Status

Calendar entries shall visually distinguish:

- Available
- Reserved
- Assigned
- Partially utilized
- Fully utilized
- Unavailable
- On break
- On leave
- Under maintenance
- Operationally blocked

- Pending approval
- Conflict detected

Utilization indicators should make it possible to immediately understand how busy a resource is.

Example:

Green: Low utilization

Yellow: Moderate utilization

Orange: High utilization

Red: Fully utilized / unavailable

Smart Actions

Selecting a free calendar slot should allow the user to:

- Assign resource
- Create reservation
- Create block
- View availability
- View resource profile

Selecting an existing assignment should allow:

- View
- Edit
- Reassign
- Extend
- Shorten
- Cancel
- Check in/out where applicable

Acceptance Condition

Authorized users can view the real-time operational status of permitted resources and initiate resource-management actions directly from the calendar.

Screen 02 — Calendar Filters, Search & Smart Discovery

Purpose

Allow users to quickly locate the correct resource instead of manually searching through hundreds or thousands of resources.

Search

Universal search shall support:

- Resource name
- Resource code
- Staff name
- Skill
- Certification
- Venue
- Room
- Category
- Equipment

- Asset
- Tag
- Event
- Experience

- **Filters**

Users shall filter by:

- Date range
- Time range
- Resource type
- Resource category
- Resource name
- Venue
- Department

- **Availability**

- Capacity
- Skill
- Certification
- Status
- Assignment status

Filters shall support combinations.

Example:

Ski Instructor + Level 3 + English + Available Tomorrow 10:00–14:00

Saved Views

Users shall be able to create saved calendar views such as:

- Ski School
- VIP Operations
- Event Resources
- Weekend Staffing
- Waterpark Rentals
- AV Equipment
- **AI Smart Search**

Conversational search should support queries such as:

- “Show available instructors tomorrow between 2 PM and 6 PM.”
- “Find a meeting room for 20 people with a projector.”
- “Show resources available for Friday's corporate event.”

Acceptance Condition

Users can identify suitable resources through structured filters or conversational AI without manually reviewing the entire resource inventory.

Screen 03 — Resource Availability Schedule Configuration

Purpose

Define the baseline schedule controlling when each resource can be used.

Configuration

Administrators shall configure:

- Available days
- Opening time
- Closing time
- Effective date
- Expiry date
- Time zone
- Seasonal availability
- Venue operating hours
- Resource-specific operating hours

Availability may be configured using:

- Standard weekly schedules
- Specific dates
- Seasonal schedules
- Exception dates
- Imported schedules
- External workforce schedules
 - **Availability Exceptions**

Administrators shall configure:

- Public holidays
- Venue closures
- Maintenance periods
- Staff leave
- Training
- Private blocks
- Operational shutdowns
- Special events

- **Inheritance**

A resource may inherit availability from:

Tenant → Venue → Department → Resource Type → Individual Resource

Individual-resource overrides shall be supported.

Acceptance Condition

Every resource has an effective availability calendar derived from configured schedules, inherited operating rules and approved exceptions.

Screen 04 — Resource Time-Slot Configuration

Purpose

Control how available resource time is converted into reservable or assignable time slots.

Configuration

Administrators shall configure:

- Slot duration

- Slot interval
- Start time
- End time
- Minimum duration
- Maximum duration
- Booking increments
- Buffer before
- Buffer after
- Capacity per slot
- Concurrent reservations
- Minimum notice
- Maximum advance booking period

Example:

- **Private Instructor**
Slot Duration: 60 minutes
Buffer: 15 minutes
Capacity: 1
- **Meeting Room**
Slot Duration: 30 minutes
Minimum Reservation: 1 hour
Maximum Reservation: 8 hours
- **Dynamic Slots**
Where enabled, TICVAI shall dynamically generate slots according to:
 - Resource availability
 - Experience duration
 - Existing bookings
 - Operating hours
 - Capacity
 - Setup requirements

Acceptance Condition

Administrators can determine precisely how resource availability is converted into customer-facing or operational booking slots.

Screen 05 — Advance Reservation Management

Purpose

Allow resources to be reserved before final operational assignment or ticket transaction.

Reservation Workspace

Users shall configure:

- Reservation reference
- Resource
- Resource type
- Date
- Start time

- End time
- Customer/event/experience
- Quantity
- Capacity
- Priority
- Reservation status
- Expiry
- Notes
- Owner

- **Reservation States**

Configurable statuses may include:

- Draft
- Tentative
- Pending Approval
- Confirmed
- Checked-In
- Completed
- Cancelled
- Expired
- No-Show

- **Reservation Holds**

TICVAI shall support temporary resource holds.

Administrators shall configure:

- Hold duration
- Automatic expiry
- Extension permissions
- Release rules

- **Smart Availability**

Before confirming a reservation, TICVAI shall automatically validate:

- **Availability**

- Capacity
- Existing bookings
- Dependencies
- Venue restrictions
- Resource status
- Operational blocks

Acceptance Condition

Authorized users can reserve resources in advance while TICVAI prevents invalid or conflicting reservations.

Screen 06 — Recurring Reservation Configuration

Purpose

Allow resources to be reserved repeatedly without creating each reservation individually.

Recurrence Patterns

Support:

- Daily
- Weekly
- Monthly
- Selected weekdays
- Selected dates
- Custom recurrence

Example:

- **Meeting Room A**

Every Monday and Wednesday

09:00–11:00

1 September–31 December

- **Recurrence Controls**

Users shall configure:

- Start date
- End date
- Frequency
- Days
- Times
- Number of occurrences
- Resource
- Venue
- Exceptions

- **Conflict Handling**

Before saving a recurring reservation, TICVAI shall analyze the entire series.

The system shall identify:

- Conflict-free occurrences
 - Conflicting occurrences
 - Unavailable dates
 - Maintenance conflicts
 - Venue closure conflicts

Users may:

- Keep resource where available
- Skip conflicting occurrences
- Select alternative resource
- Request AI optimization

Acceptance Condition

Users can create and manage recurring resource reservations while TICVAI validates every occurrence and clearly identifies exceptions.

Screen 07 — Conflict Detection & Resolution Center

Purpose

Provide a centralized workspace for detecting and resolving resource scheduling conflicts before they affect operations.

Conflict Types

The system shall identify:

- Double booking
- Capacity conflict
- Availability conflict
- Staff leave conflict
- Certification conflict
- Venue conflict
- Maintenance conflict
- Resource dependency conflict
- Setup-time conflict
- Travel-time conflict
- Parent-resource conflict
- Multi-event conflict

Conflict Severity

Conflicts may be classified:

- Informational
- Warning
- High
- Critical

Resolution Actions

Users shall be able to:

- Change time
- Change resource
- Change venue
- Reduce capacity
- Cancel assignment
- Request override
- Send for approval

AI Conflict Resolution

AI shall recommend suitable alternatives based on:

- **Availability**
- Resource type
- Skills
- Capacity

- **Location**

- Cost
- Priority
- Existing workload

Example:

Conflict detected: Alex is unavailable at 14:00.

Recommended: Maria — same certification, available 13:30–17:00, same venue.

Acceptance Condition

Scheduling conflicts are automatically identified, classified and presented with governed resolution options before operational confirmation.

Screen 08 — Operational Time & Resource Blocking

Purpose

Protect non-customer-facing time required for preparation, movement, maintenance and operational activities.

Block Types

Users shall configure:

- Setup
- Teardown
- Cleaning
- Maintenance
- Inspection
- Travel
- Break
- Training
- Private use
- Venue closure
- Operational hold
- Custom block

Configuration

Each block shall support:

- Resource
- Block type
- Start
- End
- Recurrence
- Reason
- Owner
- Priority
- Override permission
- Approval requirement

Example:

- **Event Hall A**

Setup: 14:00–16:00

Event: 16:00–20:00

Teardown: 20:00–22:00

The resource must therefore remain unavailable from **14:00–22:00**, not merely during the customer-facing event.

Acceptance Condition

Operational time blocks automatically affect resource availability and prevent reservations that would interfere with required operational activities.

Screen 09 — Multi-Event Resource Planning

Purpose

Allow operations teams to manage shared resources across multiple simultaneous events, experiences and operational activities.

Planning Workspace

The screen shall display:

- Events
- Experiences
- Resources
- Venues
- Time periods
- Assignments
- Conflicts
- Capacity
- Resource gaps

Users shall be able to compare multiple events side-by-side.

Example:

Event A

requires 10 security staff.

Event B

requires 8 security staff.

Only 14 qualified staff are available.

TICVAI shall immediately identify the **4-person staffing gap**.

Priority Management

Events may be assigned:

- Critical
- High
- Standard
- Low

Resource allocation can consider:

- Event priority
- Customer priority

- Revenue
- SLA
- Resource suitability
- Operational requirements

AI Optimization

AI should recommend how shared resources can be distributed across simultaneous activities.

Acceptance Condition

Operations teams can understand resource demand across parallel events and identify shortages or allocation conflicts before operations begin.

Screen 10 — Smart Assignment & Drag-and-Drop Reallocation

Purpose

Provide the fast, visual assignment experience that operational users and cashiers can use when assigning or changing resources.

This should become one of the signature TICVAI Resource Management experiences.

Drag-and-Drop Assignment

Users shall be able to drag:

- Reservation
- Booking
- Experience
- Staff assignment
- Resource requirement

onto an available resource and time.

The system shall immediately validate the move.

Live Validation

Before accepting the drop, TICVAI shall check:

- **Availability**
- Capacity
- Qualification
- Certification
- Dependencies
- Existing reservations
- Venue
- Travel/setup buffer
- Resource status
- Operational rules

Valid destinations should visually highlight.

Invalid destinations should visually indicate the reason.

Smart Reassignment

If an assigned resource becomes unavailable, TICVAI shall:

1. Detect the issue.
2. Identify eligible alternatives.
3. Rank alternatives.
4. Recommend the best replacement.
5. Allow authorized automatic reassignment.

Example:

Instructor Alex — Sick / Unavailable

TICVAI recommends:

1. Maria — 96% match

- Available
- Same certification
- Same venue
- No overtime

2. David — 88% match

- Available
- Qualified
- Requires venue transfer

AI Smart Assign

Users may select:

Smart Assign

TICVAI shall automatically evaluate suitable resources and recommend the best option.

The recommendation should explain **why** the resource was selected.

Customer/Cashier Flow

When a ticket or experience requires a resource, the cashier should see:

Date → Time → Available Resources → Recommended Resource → Confirm

Rather than searching manually.

The interface may show resource cards containing:

- Photo/icon
- Name
- Skill/type
- **Availability**
- Capacity
- Utilization
- Rating or suitability
- **Location**
- AI recommendation indicator

Acceptance Condition

Authorized users can assign or reassign resources visually through drag-and-drop or AI Smart Assign, with real-time validation preventing invalid allocations and all changes recorded in audit history.

Board 2 Shared Rules

All screens shall operate from a common real-time availability engine.

Changes made from one channel must immediately affect all authorized channels, including:

- Backend administration
- POS
- B2C
- B2B
- Employee App
- Event Management
- Ticketing
- API integrations

Resource availability must therefore **never be maintained independently by each channel**.

The platform shall support:

- Multi-tenant operation
- Multi-venue scheduling
- Time-zone awareness
- RBAC/PBAC
- Effective dating
- Approval workflows
- Real-time conflict detection
- API access
- Event-driven updates
- Immutable audit history
- AI recommendations

Board 2 End-to-End Outcome

Board 2 turns the resource master created in Board 1 into a **live operational scheduling engine**.

The end-to-end flow becomes:

Resource → Availability → Time Slot → Search → Reservation → Conflict Validation → Operational Block → Assignment → Smart Reassignment → Real-Time Calendar

The most important design principle is that users should not have to think:

“Which resource should I assign?”

TICVAI should understand the requirement, availability, qualifications, dependencies and operational rules and present:

“These are the best resources available. This one is recommended.”

That is the experience that makes Resource Management feel like an AI-first operational platform rather than a traditional scheduling calendar.

TICVAI Resource Management

Board 3 — Staff Resource & Workforce Configuration

Board Objective

Provide TICVAI with a centralized backend configuration environment for treating employees, instructors, contractors, technicians, security personnel, event staff, and other personnel as **intelligent operational resources**.

The screenshot displays the TICVAI Board 3 interface, which is a centralized backend configuration environment for managing staff resources. The interface is organized into a grid of ten main sections, each with a specific function:

- 01 Staff Resource Directory:** A table listing staff members with columns for Name, Role, Primary Venue, Status, and Availability. It includes filters for search and status.
- 02 Staff Resource Profile:** A detailed view of a staff member's profile, including their name, role, and various operational metrics like Skills, Certifications, and Availability.
- 03 Skills & Competency Management:** A section for managing staff skills and competencies, including a list of skills and their proficiency levels.
- 04 Certification & Expiry Management:** A table for tracking staff certifications, including the certification name, expiry date, and status.
- 05 Qualification & Assignment Rule Engine:** A section for defining rules for staff qualification and assignment, including a list of rules and their details.
- 06 Staff Availability & Working Pattern:** A calendar view showing staff availability and working patterns over time.
- 07 Shift Template & Assignment Configuration:** A section for configuring shift templates and assignments, including a list of templates and their details.
- 08 Break, Leave & Absence Configuration:** A section for configuring break, leave, and absence rules, including a list of rules and their details.
- 09 Overtime & Working-Hour Rules:** A section for configuring overtime and working-hour rules, including a list of rules and their details.
- 10 Workforce Integration & Synchronization Center:** A section for integrating and synchronizing staff data with external systems, including a list of connected systems and their status.

The board shall establish:

Staff Profile → Skills → Certifications → Qualifications → Availability → Shifts → Breaks/Leave → Overtime → External Workforce Integration

The objective is not to replace the later operational rostering board. Board 3 defines **who the employee is, what they are qualified to do, and when they are allowed or available to work.**

Operational roster construction, attendance monitoring, shift swaps, staffing gaps, and labor-cost management will remain in **Board 4** to avoid duplication.

Primary matrix coverage: 1.2.3, 1.2.5, 1.2.17, 1.2.32–1.2.38.

Supporting coverage: 1.2.4, 1.2.11–1.2.12, 1.2.34–1.2.39, 1.2.54, 1.2.60–1.2.66, 1.2.80–1.2.84.

Screen 01 — Staff Resource Directory

Purpose

Provide the centralized master directory for all personnel who may be assigned as operational resources within TICVAI.

Staff Types

The system shall support personnel such as:

- Instructors
- Event managers
- Ushers
- Security personnel
- Technical crew
- Hosts
- Performers
- Mascots
- Lifeguards
- Operators
- Customer service staff
- Cashiers
- Supervisors
- Contractors
- Temporary staff
- Customer-defined personnel types

- **Directory Information**

The directory shall display configurable columns including:

- Employee/resource ID
- Name
- Photograph
- Staff type
- Role
- Department
- Primary venue
- Skill level

- **Availability**

- Current status
- Certification status
- Shift status
- Employment type

Smart Indicators

TICVAI shall visually identify:

- Available
- Assigned
- Off shift
- On leave
- Certification expiring
- Qualification issue
- Suspended
- Unavailable

Search & Filters

Users shall filter by:

- Venue
- Department
- Role
- Staff type
- Skill
- Certification
- Qualification
- **Availability**
 - Employment type
 - Status

AI Search

Users should be able to request:

“Show all Level 3 ski instructors available this weekend.”

or:

“Find Arabic-speaking instructors qualified for private lessons.”

Acceptance Condition

Authorized users can locate and manage staff resources from one centralized directory using structured filters or conversational AI.

Screen 02 — Staff Resource Profile

Purpose

Maintain the operational resource profile for an individual employee or contractor.

This profile shall complement the organization's HR employee record rather than unnecessarily duplicate an HR system.

Identity

The screen shall support:

- Employee/resource ID
- Employee number
- Name
- Photograph
- Staff type
- Job title
- Department
- Employment type
- Primary venue
- Secondary venues
- Manager
- Operational status
 - **Operational Information**

Configure:
- Resource priority

- Customer selectable yes/no
- Individual selection allowed yes/no
- Bookable yes/no
- Scheduling enabled yes/no
- Maximum concurrent assignment
- Default assignment duration
- Preferred operating area
- Languages
- Tags

- **Employment Classification**

Support:

- Full-time
- Part-time
- Contractor
- Temporary
- Seasonal
- External agency

- **System Links**

Where an HR/workforce system exists, TICVAI shall retain:

- External employee ID
- Source system
- Synchronization status
- Last synchronization
- Record ownership

Operational Summary

The profile should provide a digital summary of:

Availability | Skills | Certifications | Upcoming Assignments | Venue Access | Current Status

Acceptance Condition

Each staff resource has a complete operational profile that can be safely referenced by scheduling, resource allocation, event management, mobile operations, and AI services.

Screen 03 — Skills & Competency Management

Purpose

Define what each staff resource is capable of performing.

Skill Library

Administrators shall create reusable skills such as:

- Ski instruction
- Snowboarding
- First aid
- Lifeguarding
- AV operation
- Lighting operation

- Crowd management
- Customer service
- Equipment operation
- Language skills

Skill Configuration

Each skill shall support:

- Skill code
- Skill name
- Skill category
- Description
- Level structure
- Applicable roles
- Applicable venues
- Applicable experiences
- Effective dates
- Active/inactive

- **Proficiency Levels**

Configurable levels may include:

- Beginner
- Intermediate
- Advanced
- Expert

or numerical structures such as:

Level 1 → Level 2 → Level 3 → Level 4

- **Employee Skill Assignment**

For each employee:

- Skill
- Proficiency level
- Effective date
- Expiry/review date
- Assessor
- Evidence
- Notes
- Verified status

- **AI Skill Matching**

The allocation engine shall use skill profiles when ranking staff for assignments.

Example:

Private Advanced Ski Lesson

Required:

- Ski Instruction: Level 3+
- English
- Private Lesson Qualified

TICVAI shall exclude personnel who fail mandatory requirements.

Acceptance Condition

Administrators can maintain reusable staff competencies and proficiency levels that are available to the assignment and AI recommendation engines.

Screen 04 — Certification & Expiry Management

Purpose

Manage certifications, licenses and credentials required for staff assignments.

Certification Library

Administrators shall configure:

- Certification code
- Certification name
- Issuing authority
- Applicable roles
- Applicable resource types
- Applicable venues
- Validity period
- Renewal requirements
- Mandatory/optional
- Warning period

Examples:

- First Aid
- Lifeguard Certification
- Instructor License
- Security License
- Equipment Operator License
- Child Safeguarding
- Health & Safety

- **Employee Certification Record**

For each employee:

- Certification
- Certificate number
- Issue date
- Expiry date
- Issuing authority
- Attachment
- Verification status
- Verified by
- Verification date

- **Expiry Management**

Configurable alerts:

- 90 days
- 60 days
- 30 days
- 14 days

- 7 days
- Expired

Operational Impact

Administrators shall configure whether expiration:

- Warns only
- Prevents new assignments
- Cancels future assignments
- Requires manager override

Acceptance Condition

TICVAI maintains current certification records, proactively identifies expiry risks and prevents unqualified staff from being assigned where mandatory certification rules apply.

Screen 05 — Qualification & Assignment Rule Engine

Purpose

Convert skills, certifications and operational policies into machine-enforceable assignment rules.

Rule Builder

Administrators shall define requirements based on:

- Role
- Skill
- Skill level
- Certification
- Language
- Age/experience where permitted
- Venue
- Department
- Experience
- Attraction
- Event type
- Resource type
- Working status

Rule Logic

Support:

- AND
- OR
- NOT
- Minimum requirement
- Preferred requirement
- Mandatory requirement

Example:

- **Advanced Private Ski Lesson**

Staff must have:

- **Ski Instructor = Yes**
- AND
- **Ski Level ≥ 3**
- AND
- **Instructor Certification = Valid**
- AND
- **First Aid = Valid**

Preferred:

- **Customer language match**
- **Qualification Result**

TICVAI shall return:

- Qualified
- Qualified with warning
- Preferred
- Not qualified

with explanation.

Rule Simulation

Administrators shall be able to test a rule before activation.

Example:

Test this rule against all instructors.

Result:

38 Qualified

7 Qualified with Warning

12 Not Qualified

Acceptance Condition

Staff assignment requirements are machine-readable, testable and automatically enforced before assignment confirmation.

Screen 06 — Staff Availability & Working Pattern

Purpose

Define when an individual staff resource is normally available to work.

Working Pattern

Configure:

- Working days
- Start time
- End time
- Time zone
- Effective period
- Weekly hours
- Availability type
- Seasonal pattern

- **Availability Types**

Support:

- Regular
- Flexible
- Seasonal
- On-call
- Temporary
- External schedule
 - **Multiple Working Patterns**

Example:

- **Peak Season**

Monday–Friday

08:00–18:00

- **Off Season**

Monday–Thursday

09:00–16:00

- **Availability Overrides**

Users shall configure:

- Additional availability
- Temporary unavailability
- Special working day
- Venue reassignment
- Restricted hours

- **Calendar Preview**

The screen should provide a live calendar preview showing:

- Available
- Working
- Unavailable
- Leave
- Break
- Existing assignment

Acceptance Condition

Each staff resource has a reliable working-availability pattern that feeds the centralized Resource Calendar and assignment engine.

Screen 07 — Shift Template & Assignment Configuration

Purpose

Configure reusable workforce shift structures that can later be used by the operational roster engine.

Shift Templates

Examples:

Morning Shift

08:00–16:00

Evening Shift

16:00–00:00

Event Shift

14:00–23:00

Split Shift

08:00–12:00 / 16:00–20:00

Shift Configuration

Each template shall support:

- Shift code
- Shift name
- Start/end time
- Duration
- Venue
- Department
- Applicable roles
- Break policy
- Overtime eligibility
- Minimum rest period
- Effective dates
 - **Assignment Controls**
Administrators shall define:
 - Maximum shifts per day
 - Maximum consecutive days
 - Minimum rest between shifts
 - Allowed roles
 - Venue restrictions
 - Skill requirements

Shift Preview

The screen shall show how the shift template will appear within the Resource Calendar and roster.

Board Boundary

This screen configures **shift templates and assignment rules**.

Actual roster construction and operational staffing will be managed in Board 4.

Acceptance Condition

Administrators can create reusable governed shift structures that can subsequently be consumed by rostering and workforce scheduling.

Screen 08 — Break, Leave & Absence Configuration**Purpose**

Ensure employee availability reflects breaks, approved leave, absences and other workforce exceptions.

Break Policies

Configure:

- Break type
- Minimum shift duration
- Break duration
- Paid/unpaid
- Mandatory/optional
- Earliest break
- Latest break
- Number of breaks

- **Leave Types**

Examples:

- Annual leave
- Sick leave
- Emergency leave
- Unpaid leave
- Training
- Personal leave
- Other configurable types

- **Leave Integration**

TICVAI shall support:

- Manual leave entry
- Approved leave imported from HR
- Workforce management integration

- **Scheduling Impact**

Approved leave shall automatically make affected staff unavailable for assignment.

- **Absence Handling**

Support:

- Planned absence
- Unplanned absence
- Sick
- No-show
- Training
- Administrative block

Where an absence impacts future assignments, TICVAI should generate a resource conflict.

Smart Replacement Trigger

When appropriate, the system shall trigger the later Smart Assignment engine:

"3 assignments affected by Maria's absence."

"Suitable replacements available."

Acceptance Condition

Breaks, leave and absences are automatically reflected in staff availability and cannot be ignored by downstream scheduling without authorized override.

Screen 09 — Overtime & Working-Hour Rules

Purpose

Configure workforce rules controlling employee working hours and overtime eligibility before operational rostering occurs.

Working-Hour Rules

Administrators shall configure:

- Standard daily hours
- Standard weekly hours
- Maximum daily hours
- Maximum weekly hours
- Minimum rest period
- Maximum consecutive working days
- Overtime threshold

Overtime Configuration

Support:

- Overtime allowed
- Overtime prohibited
- Approval required
- Maximum overtime
- Venue-specific rules
- Role-specific rules
- Employee-group rules
 - **Scheduling Warnings**
TICVAI shall detect:
 - Approaching overtime
 - Overtime generated
 - Maximum hours exceeded
 - Rest-period violation
 - Consecutive-day violation

AI Consideration

AI Smart Assignment shall consider overtime when ranking candidates.

Example:

Maria

Available

Qualified

Would create 2 hours overtime

David

Available

Qualified

No overtime

Recommendation:

David

Board Boundary

Detailed labor-cost calculations and workforce compliance monitoring remain part of Board 4.

Acceptance Condition

Working-hour and overtime rules are configurable and automatically considered before staff assignments are confirmed.

Screen 10 — Workforce Integration & Synchronization Center

Purpose

Connect TICVAI Resource Management with external HR, workforce management, payroll, identity, and employee systems where applicable.

Integration Sources

The architecture shall support integration with systems such as:

- HRMS
- Workforce Management
- Payroll
- Time & Attendance
- Identity Management
- External staffing agencies

- **Data Mapping**

Administrators shall map external fields to TICVAI including:

- Employee ID
- Name
- Department
- Role
- Venue
- Employment status
- Working hours
- Leave
- Shift
- Skills
- Certifications

Ownership Rules

For every field, administrators shall determine:

TICVAI Master

or

External System Master

This prevents conflicting updates.

Synchronization

Support:

- API
- Webhooks/event-driven integration
- Scheduled synchronization

- Manual synchronization
- File import where required
 - **Integration Monitoring**

Display:

- Connected systems
- Last synchronization
- Successful records
- Failed records
- Warnings
- Mapping errors
- Authentication status

- **Conflict Handling**

Examples:

Employee exists in HR but not TICVAI.

Employee venue changed externally.

Certification expired.

Employee terminated externally but has future TICVAI assignments.

The system shall flag affected downstream assignments.

Acceptance Condition

TICVAI can safely consume and synchronize workforce data from external systems while maintaining clear field ownership, traceability and operational consistency.

Board 3 Shared Intelligence Rules

Staff selection throughout TICVAI shall use a common **Staff Eligibility Engine**.

When evaluating an employee for an assignment, the engine should evaluate:

Availability + Shift + Skills + Certifications + Qualification + Leave + Breaks + Venue + Working Hours + Overtime + Assignment Conflicts

The result shall be explainable.

For example:

Maria Garcia — 97% Match

- ✓ Available
- ✓ Level 3 Instructor
- ✓ Certification valid
- ✓ Correct venue
- ✓ No overtime
- ✓ Language match

David Lee — 82% Match

- ✓ Available
- ✓ Qualified
- ✓ Certification valid
- ⚠ Different venue
- ⚠ 30-minute transfer required

Alex Johnson — Not Eligible

- ✗ On approved leave

AI must never bypass mandatory qualification, certification, safety, or workforce rules merely because a candidate has a high recommendation score.

Board 3 Shared Governance

All ten screens shall support, where applicable:

- Multi-tenant configuration
- Multi-venue workforce management
- RBAC/PBAC
- Maker-checker approval
- Effective dating
- Version control
- Audit history
- Arabic/English localization
- API accessibility
- Event-driven synchronization
- Mobile integration
- AI-assisted configuration

Sensitive staff information must be permission-controlled and separated from general operational resource data.

Board 3 End-to-End Outcome

After Board 3, TICVAI understands much more than simply:

“Maria is a staff resource.”

The platform understands:

Who Maria is → Where she can work → When she can work → What she can do → What she is certified to do → Which assignments she qualifies for → Whether she is on leave → Whether the assignment creates overtime

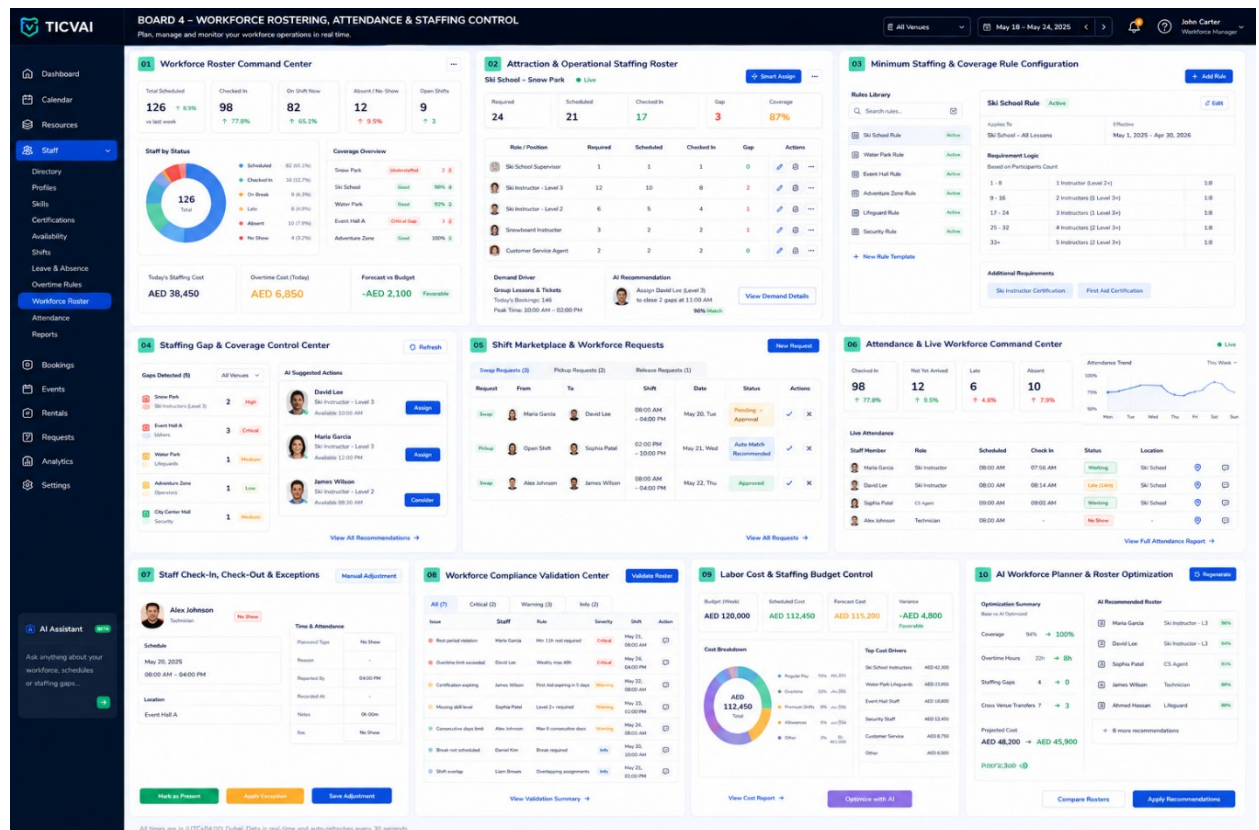
This prepares the data required for **Board 4 — Workforce Rostering, Attendance & Staffing Control**, where TICVAI will convert these rules and staff profiles into actual operational rosters, attendance monitoring, shift changes, staffing-gap detection and workforce control.

TICVAI Resource Management

Board 4 — Workforce Rostering, Attendance & Staffing Control

Board Objective

Provide TICVAI with a centralized, intelligent **Workforce Operations & Rostering Engine** that converts staff profiles, availability, skills, certifications, working-hour rules, and demand forecasts into executable operational rosters.



Board 4 shall enable managers to:

- Build operational rosters
- Assign employees to attractions, venues, events, and activities
- Define minimum staffing requirements
- Validate role and skill coverage
- Identify staffing shortages
- Manage shift swaps, transfers, pickups, and releases
- Monitor planned versus actual attendance
- Capture check-in/check-out and attendance exceptions
- Validate workforce compliance before assignments
- Calculate and forecast labor costs
- Use AI to optimize staffing

Board 3 answers:

“Who can work?”

Board 4 answers:

“Who should work, where should they work, when should they work, did they actually work, and what does that staffing plan cost?”

Primary matrix coverage: 1.2.11, 1.2.32–1.2.33, 1.2.37–1.2.39, 1.2.80–1.2.84.

Supporting coverage: 1.2.3–1.2.5, 1.2.12–1.2.13, 1.2.17, 1.2.36, 1.2.52–1.2.57, 1.2.60–1.2.66, 1.2.69, 1.2.75–1.2.76.

Screen 01 — Workforce Roster Command Center

Purpose

Provide managers with the primary operational workspace for viewing and managing workforce deployment across venues, attractions, events, departments, and shifts.

Dashboard

The command center shall display:

- Total scheduled staff
- Staff currently on shift
- Staff checked in
- Staff absent
- Late arrivals
- Open shifts
- Unfilled positions
- Understaffed operations
- Overtime risk
- Compliance warnings
- Staffing cost today
- AI staffing recommendations
 - **Roster Views**
Users shall switch between:
 - Day
 - Week
 - Month
 - Venue
 - Attraction
 - Event
 - Department
 - Employee
 - Role

Visual Timeline

The roster shall visually display:

Employee → Shift → Venue → Assignment → Break → Status

Example:

- **Maria Garcia**
08:00–10:00 | Group Ski Lesson
10:00–10:30 | Break

10:30–12:30 | Private Lesson

12:30–14:00 | Available

14:00–16:00 | Group Lesson

Smart Status

Employees shall display statuses such as:

- Scheduled
- Checked In
- Working
- On Break
- Available
- Late
- Absent
- No-Show
- Overtime
- Checked Out

- **Quick Actions**

Managers shall be able to:

- Assign employee
- Move assignment
- Add shift
- Replace employee
- Open shift
- Publish roster
- Notify employee
- Run AI optimization

Acceptance Condition

Managers can understand the complete workforce position for their permitted operations and manage staffing directly from one operational roster workspace.

Screen 02 — Attraction & Operational Staffing Roster

Purpose

Create detailed staffing plans for individual attractions, experiences, venues, departments, and events.

Roster Structure

Managers shall select:

- Date
- Venue
- Attraction
- Event
- Department
- Operating period

The system shall display required operational roles.

Example:

Ski School — 25 August

Required

- 1 Supervisor
- 6 Ski Instructors
- 2 Snowboard Instructors
- 1 Customer Service Agent

Scheduled

- 1 Supervisor ✓
- 5 Ski Instructors △
- 2 Snowboard Instructors ✓
- 1 Customer Service Agent ✓

Gap: 1 Ski Instructor

Assignment

Managers may assign employees through:

- Manual selection
- Drag and drop
- Smart Assign
- AI Auto-Roster

Employee Cards

Each candidate shall show:

- Name
- Photograph
- Role
- Skill level
- Certifications
- **Availability**
- Current hours
- Venue
- Overtime impact
- AI suitability score

Ticket/Experience Visibility

Where relevant, the roster shall show the tickets or experiences driving workforce requirements.

Example:

14:00 Group Lesson

18 tickets sold

Required instructors: 3

Assigned instructors: 2

Gap: 1

Acceptance Condition

Managers can construct operational staffing rosters against actual attraction, event, experience, and ticketing demand.

Screen 03 — Minimum Staffing & Coverage Rule Configuration

Purpose

Define the minimum personnel required to operate safely and effectively.

Rule Builder

Administrators shall configure staffing rules based on:

- Venue
- Attraction
- Experience
- Event
- Department
- Resource type
- Role
- Time
- Day
- Demand
- Capacity
- Tickets sold
- Attendance forecast

Fixed Rules

Example:

Waterpark Zone A

Minimum:

- 1 Supervisor
- 4 Lifeguards
- 1 First Aid Qualified Employee

Demand-Based Rules

Example:

Ski School

1 instructor per 8 participants.

If:

40 participants booked

TICVAI calculates:

5 instructors required

Coverage Requirements

Rules may require:

- Specific role
- Skill level
- Certification
- Language

- Supervisor presence
- First aid coverage
- Security coverage

Thresholds

Configure:

- Minimum
- Target
- Recommended
- Maximum
 - **Enforcement**

A rule may:

- Warn
- Require approval
- Prevent roster publication
- Trigger escalation

Acceptance Condition

Administrators can define fixed and demand-driven staffing requirements that TICVAI automatically evaluates against planned rosters.

Screen 04 — Staffing Gap & Coverage Control Center

Purpose

Identify workforce shortages before they create operational problems.

Gap Detection

TICVAI shall continuously compare:

Required Staffing vs Scheduled Staffing vs Actual Attendance

The system shall detect:

- Missing staff
- Missing role
- Missing skill
- Missing certification
- Understaffing
- Overstaffing
- Absence-created gap
- Demand-created gap
- Venue coverage gap

Example

Adventure Park — 14:00

Required: **12 Staff**

Scheduled: **11**

Checked In: **9**

Result:

Planned Gap: 1

Live Operational Gap: 3

Severity

Gaps may be:

- Informational
- Warning
- High
- Critical

AI Recommendations

TICVAI shall recommend:

1. Available qualified employee
2. Employee from nearby department
3. Employee from another venue
4. Shift extension
5. Approved overtime
6. External/temporary resource where configured

Smart Resolution

Example:

Critical Coverage Gap

Waterpark Zone B requires one additional certified lifeguard from 15:00–18:00.

Recommended: Ahmed Hassan — 96% Match

Acceptance Condition

Managers can identify planned and live staffing gaps and receive actionable recommendations for resolving them.

Screen 05 — Shift Marketplace & Workforce Requests

Purpose

Allow employees and managers to manage shift changes through controlled workflows rather than informal manual communication.

Supported Requests

Employees shall be able to request:

- Shift swap
- Shift transfer
- Shift pickup
- Shift release

Shift Swap

Example:

Maria

requests to exchange:

Tuesday 08:00–16:00

with:

David

Wednesday 08:00–16:00

Before allowing the swap, TICVAI shall validate:

- Role
- Skills
- Certification
- **Availability**
- Working-hour limits
- Rest periods
- Overtime
- Minimum staffing
- Venue permission

Open Shift Marketplace

Managers may publish unfilled shifts.

Eligible employees may:

- View shift
- Request pickup
- Accept shift where allowed

TICVAI shall only display shifts for which the employee meets configured eligibility rules.

Approval Workflow

Requests may require:

- Employee acceptance
- Supervisor approval
- Department approval
- Workforce manager approval

Mobile Integration

The same workflow shall be accessible from the TICVAI Employee App.

Acceptance Condition

Employees can request governed shift changes while TICVAI automatically validates qualifications, workforce rules, and operational coverage before approval.

Screen 06 — Attendance & Live Workforce Command Center

Purpose

Provide real-time visibility into whether scheduled employees actually reported and are available for operation.

Dashboard

Display:

- Scheduled today

- Checked in
- Not yet arrived
- Late
- Absent
- No-show
- On break
- Checked out
- Overtime
- Attendance exceptions

- **Planned vs Actual**

For each employee:

Employee	Planned	Actual In	Actual Out	Status
Maria	08:00	07:56	—	Working
David	08:00	08:14	—	Late
Alex	08:00	—	—	No-Show

- **Live Alerts**

Managers shall receive alerts such as:

David Lee arrived 14 minutes late.

Alex Johnson has not checked in.

Ski School staffing has fallen below minimum coverage.

- **Operational Link**

Attendance status shall immediately update:

- Resource availability
- Staffing coverage
- Assignment eligibility
- AI replacement recommendations

Acceptance Condition

Managers can monitor planned versus actual workforce attendance in real time and immediately understand operational impact.

Screen 07 — Staff Check-In, Check-Out & Attendance Exceptions

Purpose

Record actual employee working activity and manage exceptions to scheduled attendance.

Check-In Methods

TICVAI architecture shall support configurable methods such as:

- Employee App
- Manager check-in
- Staff terminal
- QR
- NFC

- Integrated time-and-attendance system

- **Attendance Record**

The system shall capture:

- Planned check-in
- Actual check-in
- Planned check-out
- Actual check-out
- Attendance status
- Lateness
- Early departure
- Overtime
- No-show
- Exception
- Source/device

- **Exception Types**

Examples:

- Late arrival
- Early departure
- Missed check-in
- Missed check-out
- Approved absence
- Sick
- Emergency
- System error
- Manager adjustment

- **Corrections**

Authorized managers may correct records.

Every correction shall capture:

- Original value
- New value
- User
- Timestamp
- Reason
- Approval where required

- **Mobile Assignment Check-In**

Where configured, employees may also check into a specific assigned resource or operational activity.

Example:

Checked In

Employee: Maria Garcia

Assignment: Private Ski Lesson

Location: Ski School Zone A

Time: 13:57

Acceptance Condition

TICVAI records actual workforce attendance and assignment activity with controlled exception handling and complete auditability.

Screen 08 — Workforce Compliance Validation Center

Purpose

Validate planned workforce schedules against legal, safety, certification, and organizational rules before roster publication or assignment.

Validation Rules

The engine shall support:

- Maximum working hours
- Mandatory breaks
- Minimum rest periods
- Overtime limits
- Consecutive working days
- Certification requirements
- Qualification requirements
- Venue rules
- Organizational workforce policies
- Applicable labor regulations
- Union rules where applicable

- **Validation Results**

Each issue shall display:

- Employee
- Rule violated
- Severity
- Affected shift
- Operational impact
- Recommended action

Example:

- **Critical**
- **Maria Garcia**
- Minimum rest required: **11 hours**
- Scheduled rest: **8 hours**

Next shift cannot be confirmed without authorized resolution.

- **Pre-Publish Validation**

Before publishing a roster, TICVAI shall run:

- **Availability Check**
- **Qualification Check**
- **Certification Check**
- **Coverage Check**
- **Working-Hour Check**
- **Rest Check**
- **Overtime Check**

- **Conflict Check**

- **Overrides**

Where policy permits an override, TICVAI shall require:

- Authorized permission
- Reason
- Approval
- Audit record

Certain critical rules may be configured as non-overridable.

Acceptance Condition

No workforce roster is published without passing configured compliance validation or following an explicitly authorized exception workflow.

Screen 09 — Labor Cost & Staffing Budget Control

Purpose

Give managers visibility into the financial impact of staffing decisions before and after roster publication.

Cost Calculation

TICVAI shall support calculation of:

- Scheduled labor cost
- Actual labor cost
- Standard hours
- Overtime cost
- Event staffing cost
- Venue staffing cost
- Attraction staffing cost
- Department staffing cost
- Employee-group cost

Budget Comparison

Display:

- **Budget**
- **Scheduled**
- **Forecast**
- **Actual**
- **Variance**

Example:

- **Event A**
- Budget: AED 18,000
- Scheduled Labor: AED 16,750
- Forecast: AED 17,400
- Variance to Budget: **AED 600 favorable**
- **Cost Drivers**

Managers shall understand:

- Base labor
- Overtime
- Additional staffing
- Temporary staff
- Cross-venue allocation
- Premium shifts

AI Cost Optimization

AI may recommend:

Replace an overtime assignment with an equally qualified available employee.

Example:

Current: Maria — AED 480 including overtime

Alternative: David — AED 320

Potential saving: AED 160

Mandatory qualification and safety rules shall always take priority over cost optimization.

Acceptance Condition

Managers can understand staffing cost and budget impact before approving workforce plans and can analyze actual labor expenditure after operations.

Screen 10 — AI Workforce Planner & Roster Optimization

Purpose

Provide TICVAI's intelligent workforce-planning experience by automatically generating or optimizing operational rosters.

AI Inputs

The engine shall consider:

- Operating schedule
- Events
- Experiences
- Ticket sales
- Forecast attendance
- Required roles
- Minimum staffing
- Skills
- Certifications
- Employee availability
- Leave
- Shift patterns
- Working-hour rules
- Rest periods
- Overtime
- Venue

- Staff preferences where permitted
- Labor costs
- Historical demand

Generate Roster

Managers may select:

Generate Roster with AI

TICVAI shall generate a recommended staffing plan.

Optimization Objectives

Administrators may configure optimization priorities such as:

- Best operational coverage
- Minimize overtime
- Minimize labor cost
- Balance employee hours
- Minimize cross-venue movement
- Maximize skill match
- Prioritize employee continuity

Safety, compliance, and mandatory qualification rules shall remain hard constraints.

- **AI Explanation**

TICVAI shall explain recommendations.

Example:

Maria Garcia — Assigned

Private Ski Lesson 14:00

Why?

- ✓ Available
- ✓ Level 3 Instructor
- ✓ Certification valid
- ✓ Customer language match
- ✓ Already at venue
- ✓ No overtime

Scenario Comparison

Managers shall compare:

Current Roster

vs.

AI Optimized Roster

Example:

Coverage: 94% → **100%**

Overtime: 22 hrs → **8 hrs**

Staffing gaps: 4 → **0**

Cross-venue transfers: 7 → **3**

Projected labor cost: AED 48,200 → **AED 45,900**

Projected Saving: AED 2,300

Human Approval

AI shall recommend or generate plans but shall not bypass configured management approval requirements.

Managers may:

- Accept all
- Accept selected recommendations
- Modify
- Reject
- Regenerate with different priorities

Acceptance Condition

TICVAI can generate and optimize workforce rosters using operational demand, employee eligibility, compliance rules, availability, and cost while retaining human governance over final publication.

Board 4 Shared Operational Logic

Board 4 shall operate from a common real-time workforce state.

A change such as:

- **Employee reports sick**

must immediately propagate through:

Attendance → Availability → Existing Assignments → Staffing Coverage → Resource Calendar → AI Replacement → Manager Alert → Employee Notification

Similarly:

Ticket demand increases

may trigger:

Demand Forecast → Staffing Requirement → Staffing Gap → AI Recommendation → Roster Adjustment

This event-driven behavior is essential to make Resource Management operationally intelligent rather than static.

Board 4 Shared Governance

All ten screens shall support, where applicable:

- Multi-tenant operation
- Multi-venue workforce control
- RBAC/PBAC
- Approval workflows
- Effective dating
- Immutable audit history
- API integration
- Event-driven integration
- Employee App integration
- Notifications
- Arabic and English
- AI explainability
- Human approval controls

All workforce decisions must remain traceable.

Board 4 End-to-End Outcome

After Board 4, TICVAI moves from knowing employee capabilities to actively managing the workforce.

The complete flow becomes:

Demand → Staffing Requirement → Roster → Staff Assignment → Coverage Validation → Employee Request → Attendance → Compliance → Labor Cost → AI Optimization

TICVAI should therefore be able to tell an operations manager:

“Tomorrow you require 126 staff across three venues. You currently have 121 eligible employees scheduled. Five coverage gaps exist. I found qualified replacements for four of them and one requires management action.”

And after optimization:

“All mandatory positions are covered. The proposed roster reduces overtime by 14 hours and projected labor cost by AED 2,100 while maintaining all qualification and compliance requirements.”

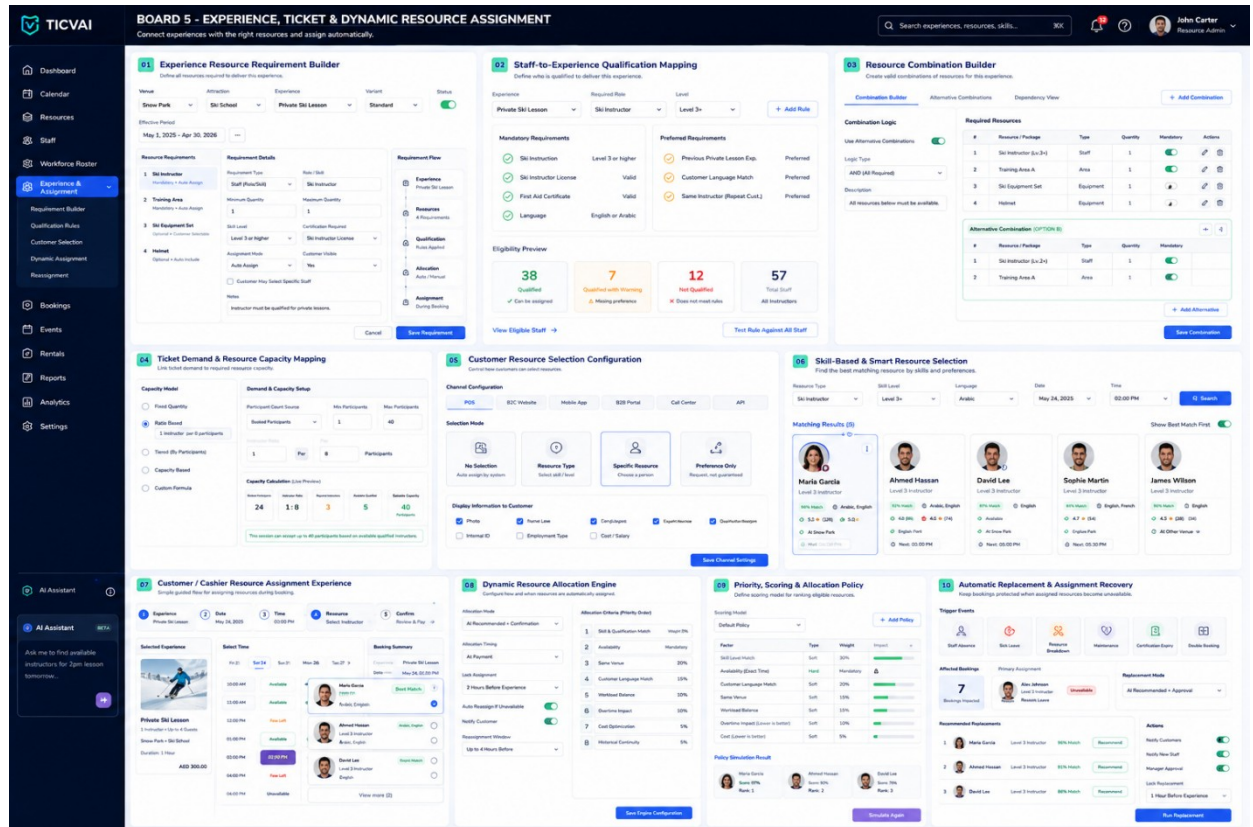
That is the intended intelligence level of the TICVAI Workforce Operations Engine.

TICVAI Resource Management

Board 5 — Experience, Ticket & Dynamic Resource Assignment

Board Objective

Provide TICVAI with a centralized **Resource Requirement & Assignment Engine** that connects ticket products, attraction experiences, sessions, and time slots with the operational resources required to deliver them.



Board 5 shall allow administrators to configure:

- Which resources an experience requires
- How many resources are required
- Which resource combinations are valid
- Which skills and qualifications are mandatory
- How resource requirements change with ticket quantity or capacity
- Whether customers may select a specific resource
- Whether customers may select a resource skill/type rather than a specific person
- Whether TICVAI should automatically allocate resources
- How resource priority is calculated
- What happens when the assigned resource becomes unavailable
- How assignments are exposed through POS, B2C, B2B, mobile, and APIs

The core principle is:

A ticket should not be considered available merely because a time slot exists. It should only be sellable when all required operational resources can also be satisfied.

Example:

A **Private Ski Lesson at 14:00** may require:

- 1 qualified ski instructor
- Level 3+ certification
- Training Area A
- Maximum 4 participants
- 15-minute preparation buffer

If the time slot exists but no qualified instructor is available, TICVAI shall understand that the experience cannot be sold unless an authorized alternative can be found.

Primary matrix coverage: 1.2.2, 1.2.4–1.2.7, 1.2.11–1.2.13, 1.2.36, 1.2.52.

Supporting coverage: 1.2.19, 1.2.28–1.2.30, 1.2.34–1.2.35, 1.2.53–1.2.57, 1.2.73–1.2.77.

Screen 01 — Experience Resource Requirement Builder

Purpose

Define which operational resources are required for each attraction, experience, ticket product, session, or service.

Experience Selection

Administrators shall select:

- Tenant
- Venue
- Attraction
- Experience
- Ticket product
- Variant
- Session type
- Effective period

Resource Requirements

For each experience, administrators shall define one or more resource requirements.

Example:

Private Ski Lesson

Required Resources

1 × Ski Instructor

1 × Training Area

Group Ski Lesson

2 × Ski Instructors

1 × Training Area

1 × Equipment Set per participant

VIP Cabana Experience

1 × Cabana

1 × Service Attendant

4 × Towels

1 × Locker

Requirement Types

Each requirement may reference:

- Specific resource
- Resource type
- Resource category
- Resource package
- Staff role
- Skill
- Certification
- Venue/space
- Equipment
- Asset
- Rental resource

Requirement Properties

Configure:

- Mandatory/optional
- Minimum quantity
- Maximum quantity
- Fixed/dynamic quantity
- Customer visible
- Customer selectable
- Auto-assign
- Allocation timing
- Priority
- Effective dates

- **Visual Requirement Map**

The screen should visually display:

- **Experience**
- ↓
- **Required Staff**
- **Required Space**
- **Required Equipment**
- **Optional Resources**

This allows administrators to understand the complete operational dependency of the experience.

Acceptance Condition

Administrators can define all operational resource requirements for an experience and TICVAI can use those requirements during availability, sale, reservation, and assignment.

Screen 02 — Staff-to-Experience Qualification Mapping

Purpose

Define which types of employees are permitted to deliver a specific experience.

Staff Requirement Configuration

Administrators shall configure:

- Required role
- Required skill
- Minimum skill level
- Required certification
- Required language
- Required venue authorization
- Experience-specific qualification
- Preferred attributes

Example:

- **Advanced Private Ski Lesson**

Mandatory:

- Role = Ski Instructor
- Ski Instruction $\geq$ Level 3
- Instructor Certification = Valid
- First Aid = Valid

Preferred:

- Same language as customer
- Previous private-lesson experience
- **Eligibility Preview**

The screen shall immediately show:

Eligible Staff: 38

Eligible with Warning: 7

Not Eligible: 12

Administrators shall be able to inspect why an employee does or does not qualify.

Qualification Source

The engine shall consume the skill, certification, and qualification rules established in Board 3 rather than recreate them.

Acceptance Condition

Every staff-based experience can be linked to machine-readable staff eligibility requirements that TICVAI automatically enforces during assignment.

Screen 03 — Resource Combination Builder

Purpose

Allow experiences to require multiple resources in valid combinations.

Combination Logic

Support:

- AND
- OR
- One of
- At least X

- Exactly X
- Optional
- Conditional

Example:

- **Corporate Event Package**

Requires:

- **1 × Meeting Room**
- AND
- **1 × AV Package**
- AND
- **1 × Technician**
- OR
- **2 × AV Operators**
- **Alternative Combinations**

Administrators may define multiple valid delivery models.

Example:

Group Lesson

Option A

2 × Level 2 Instructors

OR

Option B

1 × Level 3 Instructor

+

1 × Level 1 Instructor

Conditional Combinations

Resource combinations may change based on:

- Participant count
- Ticket type
- Customer type
- Venue
- Day
- Time
- Experience variant
- Accessibility requirement

Validation

TICVAI shall test combinations for:

- Resource availability
- Dependency rules
- Circular requirements
- Qualification
- Capacity
- Venue restrictions

Acceptance Condition

Administrators can configure flexible multi-resource combinations and TICVAI can determine whether at least one valid combination exists before confirming availability.

Screen 04 — Ticket Demand & Resource Capacity Mapping

Purpose

Connect ticket sales and participant quantity directly to resource requirements.

Capacity Models

Support:

- **Fixed**

One resource regardless of participant count.

Example:

1 Cabana per reservation.

- **Ratio-Based**

One resource per X participants.

Example:

1 Instructor per 8 participants.

- **Tiered**

1–8 participants → 1 Instructor

9–16 → 2 Instructors

17–24 → 3 Instructors

- **Capacity-Based**

Resource capacity determines maximum saleable inventory.

Example:

Training Area capacity = 30.

- **Real-Time Calculation**

TICVAI shall calculate:

Ticket Demand → Participant Count → Resource Demand → Available Resource Capacity → Saleable Capacity

Example:

Session capacity: 40

Instructor ratio: 1:8

Available qualified instructors: 4

Maximum operational capacity:

32 participants

Even if the venue capacity is 40.

Capacity Protection

The system shall prevent ticket inventory from exceeding the capacity that can actually be supported by required resources unless an authorized oversell policy exists.

Acceptance Condition

Ticket availability dynamically reflects the actual operational resource capacity required to deliver the experience.

Screen 05 — Customer Resource Selection Configuration

Purpose

Control whether customers or cashiers may choose a specific staff member or resource during purchase.

Selection Modes

Administrators shall configure:

- **No Selection**

TICVAI automatically assigns the resource.

Resource Type Selection

Customer selects:

Instructor Level 3

but not a specific instructor.

Specific Resource Selection

Customer selects:

Maria Garcia

Preference Only

Customer requests Maria, but TICVAI may assign another eligible instructor.

Channel Configuration

Selection behavior may differ by:

- POS
- B2C
- B2B
- Mobile App
- Call Center
- API

Example:

- **B2C**

Customer may choose instructor.

- **POS**

Cashier may override instructor.

- **B2B**

Auto-assignment only.

- **Resource Cards**

Selectable resources should display configurable information such as:

- Photo
- Name
- Role
- Skill
- Language

- **Availability**

- Next available time
- Venue
- Suitability
- Customer-facing description

Sensitive internal workforce data must never be exposed to customers.

Acceptance Condition

Administrators can determine precisely how resource selection works for each experience and sales channel.

Screen 06 — Skill-Based & Smart Resource Selection

Purpose

Allow users to request the type or capability of resource they require without manually choosing an individual resource.

Smart Search

Cashiers or customers may select criteria such as:

- Instructor level
- Language
- Skill
- Accessibility capability
- Equipment type
- Room capacity
- Resource feature

Example:

Ski Instructor

Level 3+

Arabic speaking

Tomorrow

14:00

TICVAI shall return only eligible resources.

Matching Score

Resources may be ranked by:

- Qualification
- Skill

- **Availability**

- Language
- Venue
- Workload
- Customer preference
- Cost
- Priority

Example:

Maria Garcia — 98% Match

- ✓ Level 3
- ✓ Arabic
- ✓ Available
- ✓ Same venue
- ✓ No overtime

David Lee — 87% Match

- ✓ Level 3
- ✗ Arabic
- ✓ Available
- ✓ Same venue

Acceptance Condition

Users can find suitable resources by operational characteristics rather than requiring knowledge of specific resource names.

Screen 07 — Customer / Cashier Resource Assignment Experience**Purpose**

Provide the fast, visual assignment interface shown during ticket or experience purchase.

This screen shall be optimized for speed and simplicity.

User Flow

The cashier/customer experience should follow:

Experience

→

Date

→

Time

→

Available Resource

→

Confirm**Calendar Experience**

After selecting a date, TICVAI shall display:

- Available times
- Limited availability
- Fully booked times
- Resource-dependent availability

After selecting a time, available resources shall appear as visual cards.

Example:

14:00 — Available Instructors**Maria Garcia**

Level 3

Arabic / English

Available

Recommended

David Lee

Level 3

English

Available

Sophie Martin

Level 2

English / French

Available

AI Recommended Option

TICVAI should highlight:

Best Match

with an explanation where appropriate.

One-Click Assignment

The cashier should be able to assign the recommended resource without opening the administrative backend.

No Resource Available

If no resource is available, TICVAI shall offer:

- Another time
- Another date
- Alternative resource
- Waiting list where configured
- Manager override where permitted

Acceptance Condition

Cashiers and authorized customer channels can assign resources through a simple visual workflow without understanding backend scheduling complexity.

Screen 08 — Dynamic Resource Allocation Engine

Purpose

Configure how TICVAI automatically chooses a resource when a customer or cashier does not make a specific selection.

Allocation Modes

Administrators shall configure:

- Manual
- Automatic
- AI recommended + user confirmation
- Fully automatic

- **Allocation Timing**

Resources may be allocated:

- During availability search
- At reservation

- At payment
- At ticket issuance
- X hours before experience
- At operational roster publication
- At customer check-in

- **Allocation Criteria**

The engine shall evaluate:

- **Availability**

- Qualification
- Skill
- Certification
- Capacity
- Venue
- Priority
- Workload
- Customer preference
- Cost
- Overtime
- Resource dependencies
- Operational continuity

Allocation Lock

Administrators shall define when an automatically assigned resource becomes locked from automatic reassignment.

Example:

Assignment locked 2 hours before experience.

API Behavior

The same allocation engine shall be available through APIs and online channels as required by the resource-management matrix.

Acceptance Condition

TICVAI can automatically allocate suitable resources consistently across backend, POS, online, mobile, and API channels.

Screen 09 — Priority, Scoring & Allocation Policy

Purpose

Configure how TICVAI ranks eligible resources when several resources could satisfy the same requirement.

Scoring Factors

Administrators may assign configurable weight to:

- Skill match
- Certification
- Customer preference

- Venue proximity
- Current workload
- Equal workload distribution
- Labor cost
- Overtime
- Resource priority
- Historical assignment
- Customer continuity
- Utilization

Example:

Skill Match — 30%

Availability — Mandatory

Customer Language — 20%

Same Venue — 15%

Workload Balance — 15%

Customer Preference — 10%

Cost — 10%

Hard vs Soft Rules

Hard Rules

Cannot be violated.

Examples:

- Mandatory certification

- **Availability**

- Safety qualification

Soft Rules

Influence ranking.

Examples:

- Language preference
- Customer preference
- Lower cost
- Same instructor as previous visit

Simulation

Administrators shall test allocation policies using sample bookings.

Example:

Booking Test

Private Ski Lesson

14:00

Arabic-speaking customer

Results:

1. Maria — 97%
2. Ahmed — 92%

3. David — 78%

The system shall explain the scoring.

Acceptance Condition

Resource allocation priorities are configurable, transparent, testable, and explainable.

Screen 10 — Automatic Replacement & Assignment Recovery

Purpose

Automatically detect when an assigned resource becomes unavailable and protect the customer's booking by finding a suitable replacement.

Trigger Events

Recovery may be triggered by:

- Staff absence
- Sick leave
- Resource breakdown
- Maintenance
- Certification expiry
- Venue closure
- Schedule change
- Double-booking resolution
- Operational emergency

- **Impact Analysis**

When a resource becomes unavailable, TICVAI shall identify:

- Affected bookings
- Affected customers
- Affected experiences
- Affected time slots
- Affected venues

Example:

Instructor Alex — Unavailable

7 future bookings affected

Replacement Search

TICVAI shall identify eligible alternatives and rank them.

Example:

Recommended Replacements

1. Maria Garcia — 96%

- ✓ Qualified
- ✓ Available
- ✓ Same venue
- ✓ Same language
- ✓ No overtime

2. David Lee — 88%

- ✓ Qualified
- ✓ Available
- △ Different venue

Recovery Modes

Administrators shall configure:

- **Automatic**
Replace without manual intervention when all rules are satisfied.
- **Approval Required**
AI recommends replacement and manager approves.
- **Manual**
Manager chooses replacement.
- **Customer Impact**
Where the replacement affects customer-facing information, TICVAI may trigger:
 - Customer notification
 - Updated ticket/booking
 - Employee notification
 - Operational notification

Audit

The system shall record:

Original Resource → Trigger → Alternatives Evaluated → Replacement → User/System Decision → Timestamp

Acceptance Condition

When an assigned resource becomes unavailable, TICVAI detects affected bookings and either automatically replaces the resource or presents governed replacement recommendations without losing the underlying customer reservation.

Board 5 Shared Resource Availability Logic

Resource-dependent products shall use a common availability calculation:

Experience Schedule

-

Ticket Capacity

-

Required Resource Availability

-

Resource Qualification

-

Resource Capacity

-

Dependencies

-

Operational Blocks

=

True Saleable Availability

A time slot shall therefore not be shown as available merely because the attraction calendar is open.

Example:

Private Ski Lesson — 14:00

Experience schedule: ✓

Training area: ✓

Ticket capacity: ✓

Qualified instructor: ✕

Result**Not Available**

or TICVAI may recommend another time where all requirements can be fulfilled.

Board 5 Shared AI Assignment Logic

TICVAI shall maintain a common **Resource Suitability Score** for eligible candidates.

Example:

Maria Garcia — 97%

- ✓ Available
- ✓ Correct skill
- ✓ Certification valid
- ✓ Same venue
- ✓ Customer language match
- ✓ No overtime
- ✓ Balanced workload

David Lee — 84%

- ✓ Available
- ✓ Qualified
- ✓ Certification valid
- ⚠ Different venue
- ⚠ Higher workload

Alex Johnson — Not Eligible

- ✕ On leave

AI recommendations must remain explainable.

Mandatory operational, safety, certification, capacity, and qualification requirements cannot be overridden by an AI recommendation score.

Board 5 Cross-Channel Architecture

The same Resource Assignment Engine shall be consumed by:

- Backend administration
- POS
- B2C website
- B2C mobile application
- B2B/reseller
- Call center
- Employee App

- Event Management
- Ticketing
- External APIs

This is critical.

TICVAI must **not create separate assignment logic for POS, B2C, and API.**

All channels shall request availability and assignment from the same centralized Resource Management service.

Board 5 Shared Governance

All screens shall support, where applicable:

- Multi-tenant architecture
- Multi-venue operation
- RBAC/PBAC
- Approval workflows
- Effective dating
- Versioning
- Immutable audit history
- API accessibility
- Event-driven integrations
- Arabic and English
- AI explainability
- Customer-data privacy
- Manual override controls
- Exception reason capture

Board 5 End-to-End Outcome

After Board 5, TICVAI understands the complete relationship between a **commercial booking and the resources required to deliver it.**

The flow becomes:

Ticket / Experience

→ **Resource Requirement**

→ **Qualification**

→ **Resource Combination**

→ **Demand & Capacity**

→ **Customer Preference**

→ **Eligible Resources**

→ **AI Ranking**

→ **Assignment**

→ **Automatic Recovery**

For the cashier, however, all this complexity should disappear.

The cashier should experience something as simple as:

Private Ski Lesson

1. Select Date

25 August

2. Select Time

14:00

3. TICVAI checks everything automatically

- ✓ Experience available
- ✓ Training area available
- ✓ Qualified instructors available
- ✓ Capacity available

4. Recommended Instructor

Maria Garcia — Best Match

5. Confirm

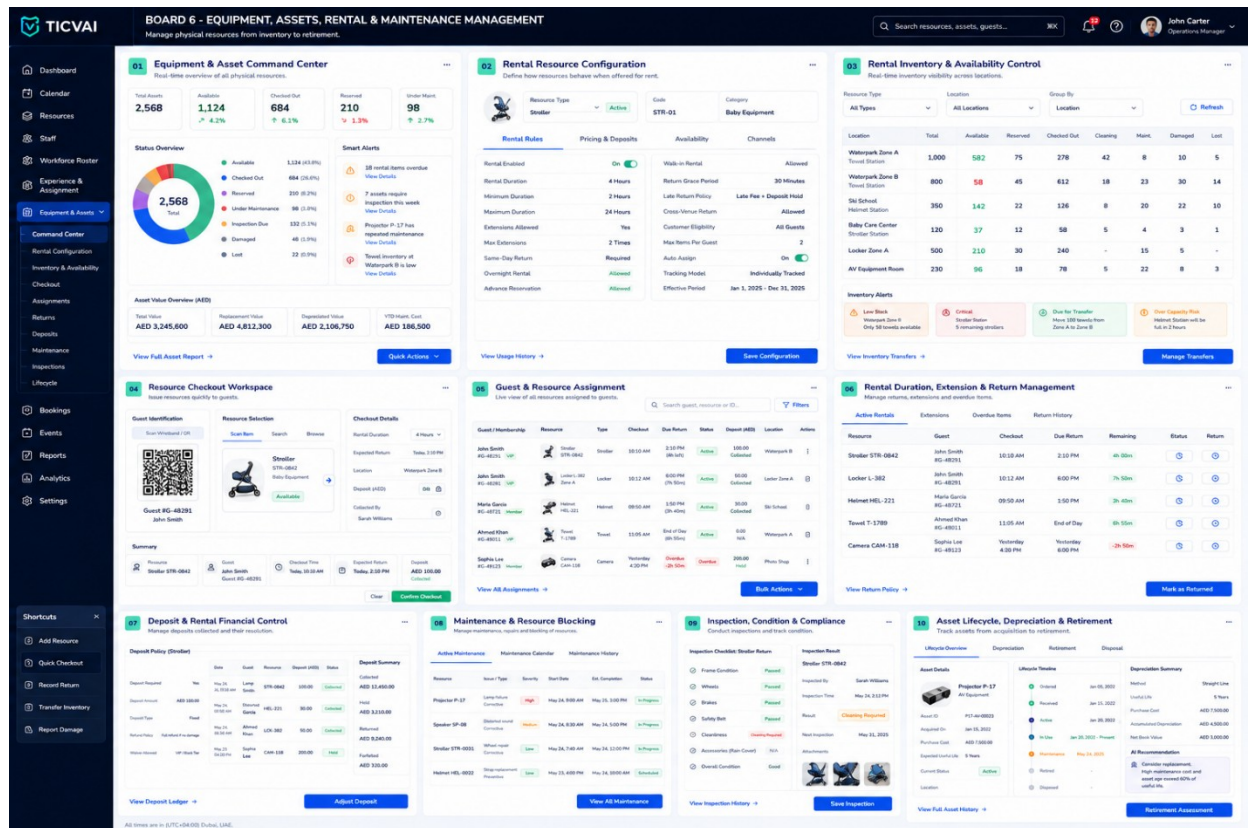
That is the objective of Board 5: **complex intelligence in the backend, extremely simple interaction in the frontend.**

TICVAI Resource Management

Board 6 — Equipment, Assets, Rental & Maintenance Management

Board Objective

Provide TICVAI with a centralized **Physical Resource & Rental Operations Engine** for managing equipment, operational assets, and customer-rental resources throughout their complete lifecycle.



Board 6 shall manage resources such as:

- Towels
- Strollers
- Wheelchairs
- Lockers
- Cabanas
- Helmets
- Sports equipment
- Cameras
- Projectors
- AV equipment
- Lighting equipment
- Sound systems
- Furniture
- Booths
- Vehicles
- Operational equipment

- Customer-defined physical assets

The board shall support:

Asset Registration → Inventory → Availability → Reservation → Checkout → Guest Assignment → Deposit → Return → Inspection → Maintenance → Depreciation → Retirement

The core objective is to give TICVAI complete visibility over:

What physical resources exist, where they are, who currently has them, what condition they are in, when they are due back, whether a deposit is held, and whether they are operationally available.

Primary matrix coverage: 1.2.8–1.2.10, 1.2.16, 1.2.40–1.2.46.

Supporting coverage: 1.2.1–1.2.2, 1.2.19, 1.2.23–1.2.30, 1.2.41–1.2.44, 1.2.67–1.2.75.

Screen 01 — Equipment & Asset Command Center

Purpose

Provide operations teams with a real-time overview of all physical resources across the organization.

Dashboard KPIs

Display:

- Total assets
- Available
- Reserved
- Checked out
- In use
- Due for return
- Overdue
- Under maintenance
- Inspection due
- Damaged
- Lost
- Retired
- **Asset Value Indicators**
 - Where financial permissions allow:
 - Total asset value
 - Replacement value
 - Depreciated value
 - Maintenance cost
 - Lost/damaged value

Resource Breakdown

Users shall view resources by:

- Resource type
- Category
- Venue
- **Location**
- Operational status

- Condition
- Ownership
- Rental status

Smart Alerts

Examples:

18 rental items overdue

7 assets require inspection this week

Projector P-17 has repeated maintenance incidents

Towel inventory at Waterpark Zone B is below threshold

Quick Actions

Users may initiate:

- Add asset
- Checkout
- Return
- Transfer
- Block for maintenance
- Record inspection
- Report damage
- Retire asset

Acceptance Condition

Authorized users can understand the real-time operational state of all physical resources and immediately identify assets requiring operational action.

Screen 02 — Rental Resource Configuration

Purpose

Define how a physical resource behaves when offered as a customer-rental resource.

Rental Configuration

Administrators shall configure:

- Rental enabled
- Rental item/resource type
- Rental duration
- Minimum duration
- Maximum duration
- Rental increments
- Same-day return required
- Overnight rental permitted
- Extension permitted
- Maximum extension
- Advance reservation permitted
- Walk-in rental permitted
- Customer eligibility

- Venue availability
 - **Quantity Model**
Support:
 - **Individually Tracked**
Example:
 - Stroller ST-00123
 - **Pooled Inventory**
Example:
 - 500 standard towels
 - **Capacity-Based**
Example:
 - 50 lockers within Locker Zone A
 - **Return Rules**
Configure:
 - Expected return time
 - Grace period
 - Late-return policy
 - Automatic overdue status
 - Return venue
 - Cross-venue return allowed
 - Inspection required
 - **Channel Rules**
Rental availability may be exposed through:
 - POS
 - B2C
 - Mobile App
 - Kiosk
 - Employee App
 - API

Acceptance Condition

Administrators can define how each physical resource or resource category behaves throughout the rental process.

Screen 03 — Rental Inventory & Availability Control

Purpose

Provide real-time inventory visibility for rental and operational resources.

Inventory Status

For each resource/category, TICVAI shall track:

- **Total Quantity**
- **Available**
- **Reserved**
- **Checked Out**
- **Under Inspection**

- **Cleaning**
- **Maintenance**
- **Damaged**
- **Lost**

Example

Waterpark Towels

Total: **1,000**

Available: **582**

Reserved: **75**

Checked Out: **278**

Cleaning: **42**

Damaged: **13**

Maintenance/Other: **10**

Location Tracking

Inventory shall be visible by:

- Venue
- Warehouse
- Store
- Rental station
- Attraction
- Zone
- Operational location

- **Inventory Thresholds**

Administrators shall configure:

- Minimum available quantity
- Reorder/transfer threshold
- Critical threshold
- Maximum quantity

- **Smart Alerts**

Example:

Towel Station B has only 58 towels available. Expected demand for the next 2 hours is 140.

Recommended:

Transfer 100 towels from Station A.

Acceptance Condition

TICVAI maintains a real-time operational inventory of rental resources and identifies shortages before they impact customer operations.

Screen 04 — Resource Checkout Workspace

Purpose

Provide frontline employees with a fast digital interface for issuing physical resources to guests.

Checkout Methods

The employee shall be able to identify the resource using:

- Search
- Barcode
- QR code
- RFID/NFC where supported
- Asset ID
- Inventory selection

Guest Identification

The resource may be linked to:

- Guest profile
- Ticket
- Booking
- Membership
- Order
- Wristband
- Mobile account

Checkout Information

Capture:

- Resource
- Guest
- Checkout time
- Expected return
- Rental duration
- Venue
- Rental station
- Employee
- Condition at checkout
- Deposit
- Notes

- **Fast Checkout**

Example:

- **Scan Guest Wristband**
→ Guest identified
- **Scan Stroller QR**
→ Resource identified
- **Deposit: AED 100**
→ Collected
- **Confirm**
→ Checked Out

The entire process should require only a few seconds.

Availability Update

Once checkout is confirmed:

Available → Checked Out

The resource shall immediately disappear from available inventory across all channels.

Acceptance Condition

Frontline employees can issue a resource to a guest through a fast scan-driven workflow, and resource availability updates immediately.

Screen 05 — Guest & Resource Assignment

Purpose

Maintain a live relationship between every checked-out physical resource and the guest responsible for it.

Assignment View

Display:

- Guest
- Photograph where available
- Ticket/booking
- Membership
- Resource
- Resource ID
- Checkout time
- Expected return
- Current status
- Deposit
- Venue
- Responsible employee
 - **Guest Resource Timeline**
Example:
 - **Guest — John Smith**
 - 10:03 — Entered Waterpark
 - 10:08 — Towel T-142 issued
 - 10:10 — Locker L-382 assigned
 - 10:14 — Stroller S-084 issued
 Expected returns:
 - Towel — Before venue closing
 - Locker — 18:00
 - Stroller — 17:30
 - **Multiple Resources**
One guest may hold multiple resources simultaneously where permitted.
 - **Restrictions**
Administrators may configure:
 - Maximum resources per guest
 - Resource-type limits
 - Eligibility requirements
 - Membership benefits
 - Deposit requirements

- **Operational Search**

Staff shall be able to search:

- “Who has stroller S-084?”
- or:
- “What resources does John Smith currently hold?”

Acceptance Condition

Every checked-out tracked resource can be traced to the responsible guest, booking, transaction, and issuing employee.

Screen 06 — Rental Duration, Extension & Return Management

Purpose

Control rental periods and manage extensions, overdue items, and returns.

Active Rental View

Display:

- Resource
- Guest
- Checkout
- Due time
- Remaining time
- Status
- Deposit
- Extension eligibility
 - **Status Indicators**
- Active
- Due Soon
- Overdue
- Extended
- Return Pending
- Returned

- **Extension**

Where permitted:

Extend Rental

TICVAI shall validate:

- Future reservation
- Operating hours
- Maximum rental duration
- Resource availability
- Additional fee
- Deposit requirement

Return Workflow

On return:

1. Identify resource.

2. Identify active rental.
3. Record return time.
4. Inspect condition.
5. Calculate additional charges.
6. Process deposit.
7. Determine next status.

Possible next status:

- Available
- Cleaning
- Inspection
- Maintenance
- Damaged

Overdue Management

TICVAI shall identify overdue resources and may trigger:

- Employee alert
- Customer notification
- Late fee
- Deposit retention workflow

Acceptance Condition

TICVAI controls the complete rental period from checkout through extension, overdue monitoring, and final return.

Screen 07 — Deposit & Rental Financial Control

Purpose

Configure and manage deposits associated with rental resources.

Deposit Configuration

Administrators shall define:

- Deposit required yes/no
- Fixed deposit
- Percentage deposit
- Deposit by resource type
- Deposit by customer type
- Deposit by membership tier
- Deposit currency
- Waiver rules

Example:

Stroller

Rental: AED 50

Deposit: AED 100

Premium Camera

Rental: AED 200

Deposit: AED 1,000

Collection Methods

The architecture shall support applicable payment methods such as:

- Cash
- Card
- Wallet
- Preauthorization
- Account balance
 - **Deposit Status**

Track:

- Required
- Collected
- Authorized
- Held
- Partially retained
- Returned
- Forfeited

- **Return Decision**

Upon resource return:

- **Good Condition**

Deposit → **Full Return**

- **Damaged**

Deposit → **Partial/Full Retention**

- **Lost**

Deposit → **Retention according to policy**

Governance

Deposit adjustments or manual release may require manager approval.

Finance Integration

All deposit movements shall be traceable to:

- Transaction
- Guest
- Resource
- Employee
- Payment reference
- Finance/ledger reference

Acceptance Condition

Deposits are collected, held, adjusted, and returned according to configurable rental policies with complete financial traceability.

Screen 08 — Maintenance & Resource Blocking

Purpose

Prevent unavailable or unsafe resources from being assigned, rented, or reserved.

Maintenance Types

Support:

- Preventive
- Corrective
- Emergency
- Scheduled
- Inspection-triggered
- Manufacturer service
 - **Maintenance Record**

Capture:

- Resource
- Maintenance type
- Issue
- Severity
- Start date/time
- Expected completion
- Actual completion
- Technician/vendor
- Cost
- Attachments
- Notes

- **Automatic Blocking**

When maintenance begins:

Available → Under Maintenance

TICVAI shall immediately prevent:

- New reservations
- Checkout
- Assignment
- Event allocation

Future Booking Impact

If future reservations exist, TICVAI shall identify them.

Example:

Projector P-01 blocked for maintenance.

3 future assignments affected.

The system should trigger the replacement engine established in Board 5.

Maintenance Calendar

Users shall see maintenance periods alongside operational availability.

Acceptance Condition

Resources under maintenance are automatically removed from operational availability and affected future assignments are identified immediately.

Screen 09 — Inspection, Condition & Compliance Management

Purpose

Ensure physical resources remain safe, compliant, and operationally fit for use.

Inspection Templates

Administrators shall configure checklists by resource type.

Example:

Stroller Return Inspection

- Frame condition
- Wheels
- Safety belt
- Brakes
- Cleanliness

AV Equipment Inspection

- Power test
- Cable condition
- Input/output test
- Physical damage
- Firmware/status

Inspection Result

Support:

- Passed
- Passed with Observation
- Cleaning Required
- Maintenance Required
- Failed
- Unsafe

- **Evidence**

Users may attach:

- Photos
- Documents
- Notes
- Signature

Compliance Scheduling

Track:

- Last inspection
- Next inspection
- Inspection frequency
- Responsible person
- Compliance status
 - **Automatic Status**

Example:

If:

- **Inspection Result = Failed**

then:

Resource Status → Blocked

and optionally:

- **Create Maintenance Request**

AI Assistance

Where enabled, AI may identify repeated failure patterns.

Example:

Projector P-17 has failed three power-related inspections in 60 days. Consider replacement assessment.

Acceptance Condition

TICVAI records inspection and condition history and automatically prevents failed or unsafe resources from operational use.

Screen 10 — Asset Lifecycle, Depreciation & Retirement

Purpose

Manage the physical resource from acquisition through operational use to final retirement.

Lifecycle

Support:

- **Ordered**
 - **Received**
 - **Active**
 - **In Use**
 - **Maintenance**
 - **Suspended**
 - **End of Life**
 - **Retired**
 - **Disposed**

- **Asset Information**

Track:

- Acquisition date
- Purchase cost
- Supplier
- Warranty
- Expected useful life
- Replacement value
- Current book/reference value
- Depreciation method
- Depreciation period

- Maintenance history
- Utilization
- Condition

- **Depreciation**

Where required, support reference tracking for:

- Straight-line depreciation
- Configurable financial method
- Accumulated depreciation
- Current value

Detailed accounting postings may remain owned by the Finance/ERP integration while Resource Management maintains operational asset references.

Retirement Assessment

TICVAI shall provide:

- Age
- Condition
- Utilization
- Maintenance cost
- Failure frequency
- Current value
- Replacement recommendation

AI Lifecycle Recommendation

Example:

Projector P-17

Age: 6.2 years

Maintenance Cost YTD: AED 4,800

Downtime: 14 days

Utilization: 71%

AI recommendation:

Consider replacement. Maintenance cost has exceeded configured lifecycle threshold.

Retirement Workflow

Retirement may require:

- Operations approval
- Finance approval
- Asset manager approval

Capture:

- Retirement reason
- Retirement date
- Disposal method
- Disposal value
- Replacement resource

Once retired, the resource must no longer appear as operationally available.

Acceptance Condition

TICVAI maintains complete lifecycle visibility for physical resources from acquisition through retirement, including operational history, maintenance, condition, value references, and governance.

Board 6 Shared Digital Resource Journey

A physical resource should have a complete digital operational history.

Example:

Stroller S-084

08:00 — Available at Rental Station A

10:14 — Checked out to Guest #48291

10:14 — AED 100 deposit collected

14:00 — Rental extended

17:22 — Returned

17:23 — Inspection: Cleaning Required

17:24 — Status changed to Cleaning

17:48 — Cleaning completed

17:49 — Available again

TICVAI therefore always knows the exact operational state of the resource.

Board 6 Shared Resource State Engine

Physical resources shall use controlled lifecycle states.

Typical operational states include:

- **Available**
- **Reserved**
- **Checked Out**
- **In Use**
- **Due Soon**
- **Overdue**
- **Returned**
- **Cleaning**
- **Inspection**
- **Maintenance**
- **Damaged**
- **Lost**
- **Suspended**
- **Retired**

Status transitions shall trigger downstream events.

Example:

Available → Checked Out

causes:

- Inventory availability reduction
- Guest-resource association
- Rental timer start

- Deposit state activation
- Audit event

Example:

Inspection Failed → Maintenance

causes:

- Resource blocking
- Future booking impact analysis
- Replacement search
- Maintenance workflow

Board 6 Shared AI Intelligence

AI shall support operational decisions without overriding safety or governance.

Examples:

- **Inventory Prediction**

"Based on tomorrow's expected attendance, Waterpark Station B may run out of towels between 13:00 and 15:00."

Transfer Recommendation

"Move 120 towels from Station A to Station B before 12:30."

Maintenance Prediction

"Projector P-17 shows an abnormal maintenance frequency."

Replacement Recommendation

"Replace Projector P-17 with P-24 for tomorrow's corporate event."

Lifecycle Recommendation

"Asset maintenance cost has exceeded the configured replacement threshold."

Board 6 Integration Requirements

Board 6 shall integrate with:

- Resource Master
- Resource Calendar
- Ticketing
- Experience Management
- POS
- B2C
- Employee App
- Event Management
- Inventory & Procurement
- Finance
- Payment services
- CRM/Guest Profile
- APIs

A resource must have **one authoritative operational state** across the TICVAI platform.

If a stroller is checked out at POS, it cannot simultaneously appear available on B2C or another POS.

Board 6 Shared Governance

All screens shall support, where applicable:

- Multi-tenant architecture
- Multi-venue operation
- RBAC/PBAC
- Approval workflows
- Effective dating
- Immutable audit history
- API access
- Event-driven integrations
- Barcode/QR/RFID identifiers
- Arabic and English
- AI explainability
- Financial traceability
- Customer-data privacy
- Mobile operation

Board 6 End-to-End Outcome

After Board 6, TICVAI can answer in real time:

What assets do we own?

Where are they?

Which ones are available?

Which guest currently has this item?

When is it due back?

Was a deposit collected?

What condition was it returned in?

Is it under maintenance?

Does it have future bookings?

What is its operational and financial lifecycle?

The complete flow becomes:

**Resource → Inventory → Availability → Reservation → Checkout → Guest Assignment → Deposit
→ Usage → Return → Inspection → Maintenance → Available Again → Lifecycle → Retirement**

The frontline experience should remain extremely simple:

Scan Guest → Scan Resource → Deposit → Confirm

and on return:

Scan Resource → Inspect → Deposit Resolution → Complete

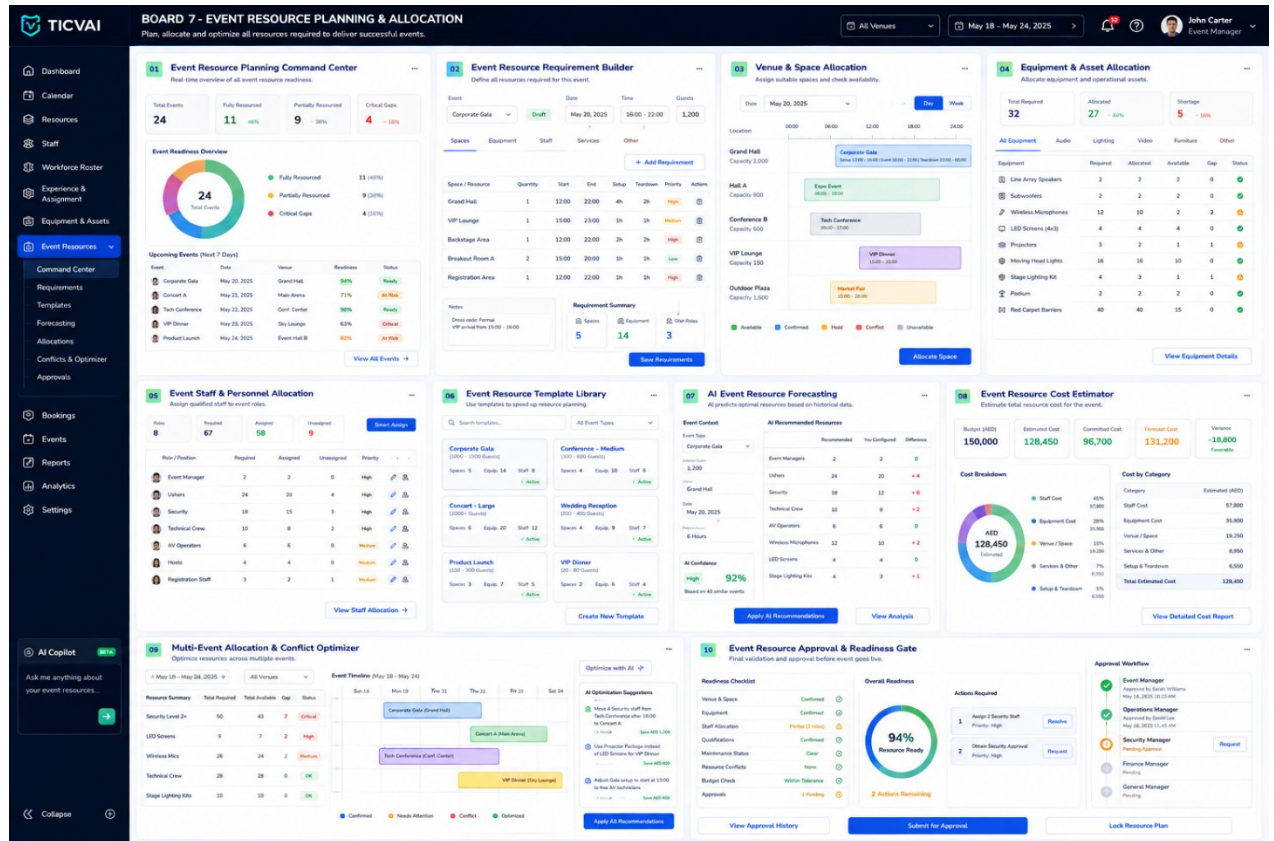
Everything else should happen intelligently in the TICVAI backend.

TICVAI Resource Management

Board 7 — Event Resource Planning & Allocation

Board Objective

Provide TICVAI with a centralized **Event Resource Planning Engine** that allows event managers and operations teams to define everything required to deliver an event and convert those requirements into actual resource reservations and assignments.



The board shall manage event requirements across:

- Venues
- Auditoriums
- Halls
- Rooms
- Stages
- Breakout areas
- Outdoor areas
- AV systems
- Lighting
- Sound systems
- Projectors
- Screens
- Chairs
- Tables
- Booths

- Operational equipment
- Event managers
- Ushers
- Security
- Technical crew
- Hosts
- Performers
- Mascots
- Temporary staff
- Other configurable resources

The event-resource lifecycle shall be:

Event → Requirements → Template → Demand Forecast → Availability → Allocation → Conflict Check → Cost → Approval → Operational Plan

The objective is to ensure that an event is not considered operationally ready merely because the venue and event schedule have been created.

TICVAI must understand:

Do we actually have all the people, spaces, equipment and operational resources required to deliver this event?

Primary matrix coverage: 1.2.14–1.2.22, 1.2.40, 1.2.47–1.2.52.

Supporting coverage: 1.2.19, 1.2.29–1.2.31, 1.2.39, 1.2.41–1.2.43, 1.2.53–1.2.58, 1.2.67–1.2.76.

Screen 01 — Event Resource Planning Command Center

Purpose

Provide event and operations managers with a single workspace for understanding the resource readiness of upcoming events.

Dashboard KPIs

Display:

- Upcoming events
- Events fully resourced
- Events partially resourced
- Events with critical gaps
- Pending resource requests
- Pending approvals
- Resource conflicts
- Staff shortages
- Equipment shortages
- Venue/space conflicts
- Forecast resource cost

- **Event Readiness**

Each event shall receive a visual readiness indicator.

Example:

Corporate Gala

Resource Readiness: 94%

- ✓ Venue confirmed
- ✓ AV confirmed
- ✓ Catering resources confirmed
- ✓ Technical crew confirmed
- ⚠ 2 security positions unfilled

Concert A**Resource Readiness: 71%**

- ✓ Venue confirmed
- ⚠ Lighting incomplete
- ✗ Sound engineer unavailable
- ✗ 8 ushers unassigned

Filters

Users shall filter by:

- Event
- Date
- Venue
- Event type
- Resource status
- Readiness
- Event manager
- Approval status
- Risk level

- **AI Event Summary**

Example:

5 events are scheduled this weekend.

3 are fully resourced.

2 have unresolved staffing or equipment gaps.

Acceptance Condition

Managers can immediately understand resource readiness across upcoming events and identify events requiring operational attention.

Screen 02 — Event Resource Requirement Builder**Purpose**

Define every resource required to deliver a specific event.

Event Selection

The screen shall consume the event master from TICVAI Event Management.

Display:

- Event
- Event type
- Venue
- Event date
- Start/end

- Expected attendance
- Capacity
- Event manager

- **Resource Requirement Sections**

- **Spaces**

Configure:

- Hall
- Auditorium
- Stage
- Breakout room
- Backstage
- VIP area
- Registration area

- **Equipment**

Configure:

- Sound
- Lighting
- Screens
- Projectors
- Microphones
- Furniture
- Barriers
- Booths

- **Staff**

Configure:

- Event managers
- Ushers
- Security
- Technical crew
- Hosts
- Performers
- Operational personnel

- **Requirement Properties**

For each requirement:

- Resource type
- Category
- Quantity
- Mandatory/optional
- Start
- End
- Setup requirement
- Teardown requirement
- Skill requirement
- Certification
- Priority
- Substitute allowed

- Approval required
 - **Dynamic Requirements**
Requirements may change according to:
- Expected attendance
- Tickets sold
- Event type
- Venue
- Duration
- VIP attendance
- Event risk classification

Acceptance Condition

Event managers can define a complete structured resource requirement plan for each event.

Screen 03 — Venue & Space Allocation

Purpose

Assign suitable physical spaces to an event and its operational components.

Space Selection

The interface shall display available:

- Venues
- Halls
- Rooms
- Stages
- Breakout areas
- Outdoor spaces
- VIP areas
- Backstage areas

Availability Timeline

Users shall see:

Setup → Event → Teardown

Example:

Grand Hall

Setup

12:00–16:00

Event

16:00–22:00

Teardown

22:00–00:00

The hall shall therefore remain operationally unavailable from:

12:00–00:00

Space Information

Display:

- Capacity
- Layout
- **Location**
- **Availability**
 - Existing bookings
 - Setup time
 - Teardown time
 - Accessibility
 - Associated resources
 - Operational restrictions

Conflict Detection

TICVAI shall prevent:

- Double booking
- Overlapping setup
- Overlapping teardown
- Maintenance conflict
- Capacity violation

Alternative Recommendation

If the requested space is unavailable:

Grand Hall unavailable.

Recommended:

Conference Hall B — 94% match

- ✓ Capacity suitable
- ✓ Available
- ✓ Same venue
- ⚠ 10-minute additional setup

Acceptance Condition

Users can assign suitable venues and spaces while TICVAI validates the entire operational occupancy period rather than only the event start/end time.

Screen 04 — Equipment & Asset Allocation

Purpose

Allocate physical equipment and operational assets required for events.

Equipment Requirements

Users shall assign:

- AV systems
- Sound systems
- Lighting
- Projectors
- Screens

- Microphones
- Cameras
- Furniture
- Booths
- Barriers
- Operational equipment

Allocation View

For each requirement, show:

Required

Allocated

Available

Gap

Example:

Resource	Required	Allocated	Gap
Projectors	4	4	0
Wireless Mics	12	10	2
LED Screens	3	3	0
Lighting Rigs	4	3	1

Asset Validation

Before allocation TICVAI shall check:

- **Availability**
- Maintenance
- Inspection status
- **Location**
- Existing reservation
- Setup requirements
- Dependencies

Specific vs Dynamic Assignment

Event managers may request:

Specific Asset

Projector P-01

or:

Resource Type

1 × 4K Projector

TICVAI may then choose the most appropriate available unit.

Acceptance Condition

Event equipment requirements can be translated into actual physical-asset allocations without creating resource conflicts.

Screen 05 — Event Staff & Personnel Allocation

Purpose

Assign qualified staff resources to event roles.

Event Roles

Support:

- Event manager
- Supervisor
- Ushers
- Security
- Technical crew
- AV operators
- Hosts
- Performers
- Mascots
- Registration staff
- Customer service
- Custom roles

Requirement Configuration

For each role:

- Required quantity
- Minimum skill
- Certification
- Shift
- Start/end
- Break requirement
- Venue
- Priority

Candidate Selection

TICVAI shall consume Board 3 and Board 4 workforce intelligence.

Each candidate may display:

- Name
- Role
- Skill
- Certification
- **Availability**
 - Existing workload
 - Venue
 - Overtime impact
 - AI match

AI Smart Assign

Example:

AV Technician Required

Maria — 97%

- ✓ AV Level 3
- ✓ Available
- ✓ Same venue
- ✓ No overtime

David — 86%

- ✓ AV Level 3
- ✓ Available
- ⚠ Different venue

Acceptance Condition

Event managers can assign qualified staff while TICVAI automatically validates workforce eligibility and availability.

Screen 06 — Event Resource Template Library

Purpose

Allow frequently used event-resource configurations to be saved as reusable templates.

Example Templates

Corporate Conference — 500 Guests

- 1 Main Hall
- 3 Breakout Rooms
- 1 Stage
- 2 LED Screens
- 8 Wireless Microphones
- 1 Sound System
- 1 Lighting Package
- 1 Event Manager
- 8 Ushers
- 6 Security
- 4 Technical Crew

Concert — Medium

- Main Stage
- Backstage
- Lighting Package
- Sound Package
- 12 Security
- 15 Ushers
- 8 Technical Crew

Template Configuration

Administrators shall configure:

- Template code
- Template name
- Event type
- Attendance range
- Applicable venues
- Required resources
- Quantities
- Staff roles
- Setup duration
- Teardown duration
- Dependencies
- Effective dates

- **Template Application**

When creating an event:

Apply Resource Template

TICVAI shall populate the resource plan automatically.

Users may then modify the event-specific requirements.

Template Versioning

Changes shall not silently modify already-approved event plans.

Acceptance Condition

Reusable event-resource templates significantly reduce manual event setup while maintaining controlled versioning.

Screen 07 — AI Event Resource Forecasting

Purpose

Predict the resources required for an event before final allocation.

Forecast Inputs

AI shall consider:

- Event type
- Expected attendance
- Tickets sold
- Venue
- Event duration
- Historical events
- Setup complexity
- VIP requirements
- Security classification
- Operational standards
- Season
- Day/time

Forecast Output

Example:

Corporate Gala — 1,200 Guests

AI Recommended:

- Security: 18
- Ushers: 24
- Event Managers: 2
- Technical Crew: 8
- Registration Staff: 6
- Wireless Microphones: 10
- LED Screens: 4
- Barriers: 20

Comparison

Display:

- **Configured**

vs.

- **AI Recommended**

Example:

Security:

- Configured: **12**
- AI Recommended: **18**
- Difference: **+6**

Reason:

Historical events of similar size required additional security during entrance and exit periods.

- **Human Control**

AI recommendations may be:

- Accepted
- Partially accepted
- Rejected
- Modified

Acceptance Condition

TICVAI can forecast event-resource demand and explain differences between manually configured and predicted requirements.

Screen 08 — Event Resource Cost Estimator

Purpose

Calculate the expected resource cost of delivering an event.

Cost Components

Support:

- Venue cost
- Room/space cost
- Equipment cost

- Rental cost
- Staff cost
- Overtime
- External resource cost
- Setup cost
- Teardown cost
- Transportation/transfer cost
 - **Cost Summary**
Display:
 - **Budget**
 - **Estimated**
 - **Committed**
 - **Actual**
 - **Variance**
Example:
 - **Corporate Gala**
 - Budget: AED 120,000
 - Estimated Resource Cost: AED 108,500
 - Committed: AED 91,200
 - Forecast Actual: AED 112,000
 - Budget Variance:
 - **AED 8,000 favorable**
 - **Cost Breakdown**
Users shall analyze by:
- Resource type
- Department
- Venue
- Event phase
- Internal/external
- Staff/equipment
 - **AI Optimization**
AI may recommend:
Replace 3 cross-venue technicians with equally qualified local staff.
Estimated saving:
 - **AED 1,850**

Acceptance Condition

Event managers can understand expected resource cost before approving the operational event plan.

Screen 09 — Multi-Event Allocation & Conflict Optimizer

Purpose

Manage scarce resources across simultaneous or overlapping events.

Planning View

Display multiple events across a shared timeline.

Example:

Friday

Corporate Gala

Concert A

Conference B

VIP Dinner

The screen shall compare resource demand across all events.

Shared Resource Analysis

Example:

Security Level 2+

Required:

Corporate Gala — 18

Concert A — 24

Conference B — 8

Total Required: 50

Available: **43**

Gap: 7

Equipment Conflict

Example:

LED Screens Required: **9**

Available: **7**

Gap: **2**

Priority Rules

Allocation may consider:

- Event priority
- SLA
- Customer tier
- Revenue
- Contractual commitment
- Resource suitability
- Operational risk

AI Optimization

AI shall propose:

- Resource redistribution
- Alternative resources
- Schedule adjustments
- Cross-venue transfers
- External resource requirement

Example:

Move 4 security staff from Conference B after 18:00 to Concert A.

Replace LED Screen requirement at VIP Dinner with compatible projector package.

Scenario Comparison

Managers shall compare:

Current Allocation

vs.

AI Optimized Allocation

including:

- Gaps
- Conflicts
- Costs
- Transfers
- Utilization

Acceptance Condition

Managers can resolve shared-resource conflicts across multiple simultaneous events using visual planning and AI-supported optimization.

Screen 10 — Event Resource Approval & Readiness Gate**Purpose**

Provide the final governance checkpoint before an event resource plan becomes operational.

Readiness Checklist

TICVAI shall validate:

Venue & Space

✓ Confirmed

Equipment

✓ Allocated

Staff

△ 1 position pending

Qualifications

✓ Valid

Maintenance

✓ Clear

Resource Conflicts

✓ None

Budget

✓ Within tolerance

Approvals

△ Security approval pending

Readiness Status

Events may be classified:

- Draft
- Planning

- Partially Resourced
- Resource Ready
- Pending Approval
- Approved
- Operationally Ready
- Blocked

Approval Workflow

Depending on event size/risk, approval may require:

- Event Manager
- Operations
- Security
- Technical
- Finance
- Venue Manager
- Senior Management

Readiness Gate

TICVAI shall prevent an event from reaching **Operationally Ready** where mandatory resource requirements remain unresolved.

Critical requirements shall be non-overridable where configured.

Publication

Once approved:

- Resource reservations are locked as configured.
- Staff assignments are published.
- Employee notifications are triggered.
- Event calendar is updated.
- Equipment allocations are confirmed.
- Operational teams receive the resource plan.

Audit

Record:

- Resource-plan version
- Submitted by
- Approvers
- Approval dates
- Exceptions
- Overrides
- Final allocation
- Publication status

Acceptance Condition

An event cannot be marked operationally ready until mandatory resource requirements, conflicts, qualifications, costs, and approvals have passed configured readiness controls.

Board 7 Shared Event Readiness Engine

TICVAI shall calculate an event's resource readiness using a common model:

Space Readiness

-

Equipment Readiness

-

Staff Readiness

-

Qualification Readiness

-

Availability

-

Conflict Status

-

Operational Timing

-

Approval Status

=

Event Resource Readiness

Example:

Corporate Gala

Venue: **100%**

Equipment: **100%**

Staff: **92%**

Qualifications: **100%**

Conflicts: **Clear**

Approvals: **80%**

Overall

94% Resource Ready

TICVAI shall clearly explain what prevents the event from reaching 100%.

Board 7 Shared AI Event Copilot

Managers should be able to interact with the event-resource plan conversationally.

Examples:

“Prepare the resources for a 1,500-person concert next Friday.”

TICVAI may respond with a proposed requirement plan based on templates, venue configuration, operational policies, and forecasts.

Or:

“Why isn't Corporate Gala resource-ready?”

TICVAI:

“Two Level 2 security positions remain unfilled and Security approval is pending.”

Or:

“Can we operate all four events Friday without external resources?”

TICVAI should simulate the scenario and identify shortages.

AI-generated plans must remain subject to mandatory operational and approval rules.

Board 7 Integration Requirements

Board 7 shall consume and coordinate information from:

- Event Management
- Resource Master
- Resource Calendar
- Workforce Management
- Equipment & Asset Management
- Ticketing
- Experience Management
- Finance
- Employee App
- Notifications
- APIs

Resource definitions must remain owned by Resource Management.

Event definitions must remain owned by Event Management.

Board 7 creates the **operational relationship between them** rather than duplicating either master.

Board 7 Shared Governance

All screens shall support, where applicable:

- Multi-tenant architecture
- Multi-venue operation
- RBAC/PBAC
- Configurable approvals
- Effective dating
- Versioning
- Immutable audit history
- Event-driven integration
- API accessibility
- Arabic and English
- AI explainability
- Cost permissions
- Override governance
- Notifications

Board 7 End-to-End Outcome

After Board 7, TICVAI should be able to take an event such as:

- **Corporate Gala**
- **1,200 Guests — Grand Hall — Friday 18:00**
and automatically understand:
- **Which spaces are required**
 - **Which equipment is required**
 - **Which staff roles are required**
 - **How many resources are needed**
 - **Which resources are actually available**
 - **Which resources are qualified**
 - **What conflicts exist**
 - **What the resources will cost**
 - **What approvals remain**
 - **Whether the event is operationally ready**

The event manager should ultimately see a simple status:

94% RESOURCE READY

2 actions remaining

- △ Assign 2 Security Staff
- △ Obtain Security Approval

TICVAI has identified qualified replacements for both open positions.

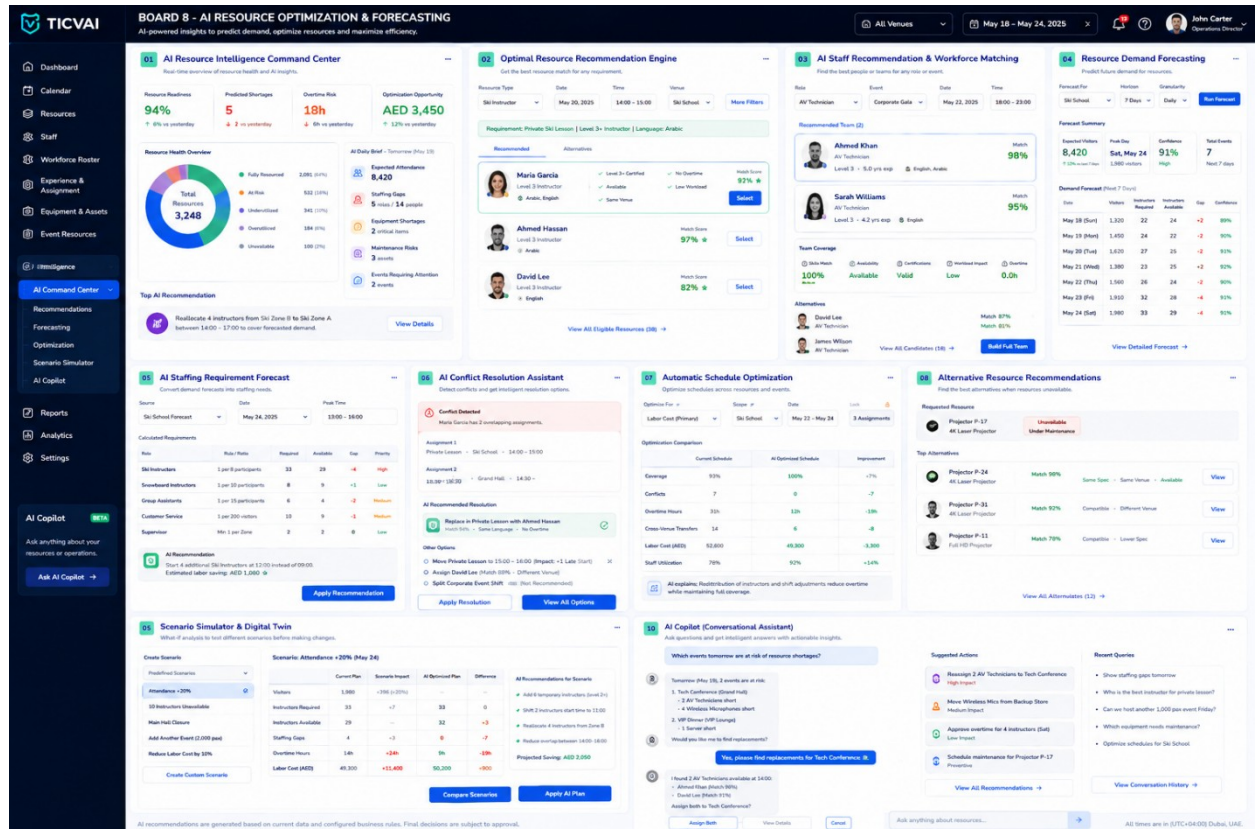
That is the intended Board 7 experience: the event manager should not need to manually check five different systems to determine whether an event can actually operate.

TICVAI Resource Management

Board 8 — AI Resource Optimization & Forecasting

Board Objective

Provide TICVAI with a centralized **AI Resource Intelligence Engine** capable of forecasting demand, identifying resource shortages, recommending optimal staff and physical resources, resolving conflicts, optimizing schedules, simulating operational scenarios, and allowing managers to interact with Resource Management through natural language.



Board 8 shall consume information from the previous Resource Management boards rather than recreate their configuration.

The AI engine shall analyze:

- Resource master data
- Resource availability
- Calendars
- Reservations
- Ticket sales
- Experience bookings
- Event schedules
- Staff skills
- Certifications
- Workforce availability
- Rosters

- Attendance
- Labor costs
- Equipment availability
- Maintenance
- Resource utilization
- Historical demand
- Venue operations
- Resource conflicts
- Customer preferences
- Operational priorities

The intelligence flow shall be:

Demand → Forecast → Required Resources → Available Resources → Eligibility → Optimization → Recommendation → Simulation → Approval → Execution

The core objective is for TICVAI to move beyond:

“Here are your resources.”

and instead tell the operator:

“Here is what you will need, where you will have shortages, and the best action to take.”

Primary matrix coverage: 1.2.39, 1.2.48, 1.2.53–1.2.59.

Supporting coverage: 1.2.11–1.2.13, 1.2.18–1.2.22, 1.2.29–1.2.31, 1.2.36, 1.2.47–1.2.52, 1.2.67–1.2.76.

Screen 01 — AI Resource Intelligence Command Center

Purpose

Provide management with a centralized AI-generated overview of current and future resource conditions across the organization.

Intelligence KPIs

Display:

- Resource readiness
- Today's utilization
- Forecast utilization
- Predicted shortages
- Predicted excess capacity
- Staffing gaps
- Equipment gaps
- Resource conflicts
- Overtime risk
- Maintenance risk
- AI recommendations awaiting action
- Estimated optimization opportunity

- **Time Horizons**

Users shall switch between:

- Live
- Today

- Tomorrow
- 7 Days
- 30 Days
- Custom period

Resource Intelligence

The dashboard shall identify:

Underutilized

Resources with significant available capacity.

Overutilized

Resources operating close to or above configured thresholds.

Shortage Risk

Resources where forecast demand exceeds available supply.

Operational Risk

Resources affected by:

- Maintenance
- Staff absence
- Certification expiry
- Scheduling conflicts
- Venue restrictions

- **AI Daily Brief**

Example:

- **TICVAI AI Brief — Tomorrow**
- **Expected Attendance:** 8,420
- **Resource Readiness:** 94%
- **5 staffing gaps predicted**
- **2 equipment shortages predicted**
- **18 overtime hours at risk**
- **AED 4,200 potential labor optimization**

Top Recommendation:

Reallocate four instructors from Ski Zone B to Ski Zone A between 14:00 and 17:00.

Acceptance Condition

Authorized managers can immediately understand current and forecast resource conditions and see prioritized AI-generated actions from one command center.

Screen 02 — Optimal Resource Recommendation Engine

Purpose

Recommend the best available resource for any operational requirement.

Request Inputs

The engine shall accept requirements such as:

- Resource type
- Date/time
- Venue

- Capacity
- Skill
- Certification
- Equipment specification
- Duration
- Customer preference
- Operational priority

- **Candidate Evaluation**

TICVAI shall evaluate eligible candidates against:

- **Availability**

- Resource type
- Skills
- Qualifications
- Certifications
- Capacity
- Venue
- Distance/transfer time
- Utilization
- Current assignments
- Cost
- Maintenance status
- Dependencies
- Customer preference

Recommendation

Example:

Requirement

Private Ski Instructor

25 August

14:00–15:00

Arabic-speaking

Level 3+

AI Recommendation

Maria Garcia — 97% Match

- ✓ Available
- ✓ Level 3
- ✓ Arabic
- ✓ Same venue
- ✓ Certification valid
- ✓ No overtime

Alternatives

Ahmed Hassan — 91%

David Lee — 82%

Explainability

Users shall be able to select:

Why this resource?

TICVAI shall explain the scoring factors.

Acceptance Condition

The AI engine can rank suitable resources for a defined operational requirement and clearly explain why each resource was recommended.

Screen 03 — AI Staff Recommendation & Workforce Matching

Purpose

Provide advanced AI-assisted selection specifically for human resources.

This screen consumes the staff rules established in Boards 3 and 4.

AI Evaluation

Staff recommendations shall consider:

- Role
- Skills
- Skill level
- Certifications
- Experience
- Language
- **Availability**
- Shift
- Leave
- Existing assignments
- Working hours
- Rest periods
- Overtime
- Venue
- Customer preference
- Workload balance
- Historical performance where permitted

Smart Candidate Ranking

Example:

Requirement

2 × AV Technicians

Event: Corporate Gala

18:00–23:00

Recommended:

1. Ahmed — 98%

2. Sarah — 95%

Alternatives:

3. David — 87%

Team Recommendation

AI shall also be able to recommend a complete team rather than individual employees.

Example:

Build the best available technical crew for Corporate Gala.

TICVAI may recommend:

- 1 Technical Supervisor
- 2 AV Technicians
- 2 Lighting Technicians
- 1 Sound Engineer

Fairness & Governance

Where workforce recommendations use performance, preferences, or historical information, administrators shall control which factors are permitted.

Mandatory qualifications and workforce rules shall always remain hard constraints.

Acceptance Condition

TICVAI can recommend individual staff members or complete qualified teams while respecting availability, compliance, skills, and workforce governance.

Screen 04 — Resource Demand Forecasting

Purpose

Predict future demand for resources before shortages occur.

Forecast Inputs

The engine shall consider available data such as:

- Historical attendance
- Ticket sales
- Advance bookings
- Event schedule
- Experience reservations
- Season
- Day of week
- Time
- Venue
- Holidays
- Campaigns
- Historical utilization
- Current booking pace

External forecast data may be incorporated through approved integrations where configured.

- **Forecast Levels**

Forecast demand by:

- Tenant
- Venue
- Attraction

- Event
- Resource type
- Resource category
- Time slot

Example

Saturday — Ski School

Expected Visitors:

2,850

Expected Lesson Participants:

620

Forecast Instructor Requirement:

48

Available:

42

Predicted Gap

6 instructors

Confidence

Forecasts should display confidence indicators where technically supported.

Example:

Forecast Confidence: 91%

Comparison

Display:

- **Historical**
vs.
- **Forecast**
vs.
- **Actual**
to improve forecasting evaluation.

Acceptance Condition

TICVAI can forecast future resource demand at useful operational levels and identify likely shortages or excess capacity before the operating date.

Screen 05 — AI Staffing Requirement Forecast

Purpose

Translate forecast operational demand into specific workforce requirements.

Demand Conversion

TICVAI shall use:

- Minimum staffing rules
- Participant ratios
- Venue requirements

- Experience rules
- Event requirements
- Safety requirements
- Expected attendance

Example:

Waterpark

Forecast Attendance:

5,200

Configured rule:

1 Lifeguard / 250 guests

Minimum operational base:

14 Lifeguards

Calculated requirement:

21 Lifeguards

Available scheduled:

18

Gap

3 Lifeguards

Time-Based Forecast

Requirements shall be forecast by operational interval.

Example:

09:00 → 14 Lifeguards

11:00 → 17

13:00 → 21

15:00 → 22

17:00 → 18

This prevents overstaffing the entire day based only on peak demand.

AI Recommendation

TICVAI may recommend:

Start three additional lifeguards at 12:00 rather than 09:00.

Estimated labor saving:

AED 1,080

Acceptance Condition

Forecast demand is automatically translated into role, skill, certification, quantity, and time-specific workforce requirements.

Screen 06 — AI Conflict Resolution Assistant

Purpose

Automatically analyze resource conflicts and recommend the most operationally appropriate resolution.

Conflict Inputs

Consume conflicts from:

- Resource Calendar
- Workforce Roster
- Events
- Equipment
- Reservations
- Experience assignments

Conflict Types

Support:

- Double booking
- Staff absence
- Qualification failure
- Certification expiry
- Equipment maintenance
- Venue conflict
- Capacity conflict
- Dependency conflict
- Travel-time conflict
- Setup conflict
- Multi-event resource shortage

- **Resolution Recommendations**

For each conflict, AI shall evaluate alternatives such as:

- Replace resource
- Move assignment
- Change time
- Change venue
- Redistribute workload
- Use substitute resource
- Transfer resource
- Request external resource
- Request controlled override where allowed

Example**Conflict****Maria Garcia**

Assigned:

Private Lesson — 14:00

Corporate Event — 14:30

AI Recommendation**Move Private Lesson to Ahmed Hassan**

Match: 94%

Impact:

- ✓ Customer language retained
- ✓ No overtime
- ✓ No downstream conflict

Bulk Resolution

Managers shall be able to select:

Resolve All Recommended Conflicts

subject to approval policies.

Acceptance Condition

AI can analyze operational conflicts, identify viable alternatives, rank resolutions, and explain the expected operational impact before execution.

Screen 07 — Automatic Schedule Optimization

Purpose

Optimize resource schedules across multiple assignments while respecting operational constraints.

Optimization Scope

Users shall select:

- Venue
- Department
- Event
- Attraction
- Resource category
- Date range

Optimization Objectives

Administrators may prioritize:

- Maximize coverage
- Minimize conflicts
- Minimize overtime
- Minimize cost
- Balance workload
- Maximize utilization
- Minimize idle time
- Minimize cross-venue movement
- Maximize customer preference match

Hard Constraints

Optimization shall never violate configured mandatory rules such as:

- Certification
- Qualification
- Safety
- Capacity

- **Availability**

- Required rest
- Resource dependency

Before vs Optimized

Example:

Current Schedule

Coverage: **93%**

Conflicts: **7**

Overtime: **31 hrs**

Transfers: **14**

Labor Cost: **AED 52,600**

AI Optimized

Coverage: **100%**

Conflicts: **0**

Overtime: **12 hrs**

Transfers: **6**

Labor Cost: **AED 49,300**

Improvement

AED 3,300 projected saving

Human Control

Users may:

- Preview
- Accept all
- Accept selected
- Reject
- Lock assignments
- Re-run optimization

Acceptance Condition

TICVAI can generate an improved resource schedule while respecting mandatory operational rules and maintaining human approval over execution.

Screen 08 — Alternative & Replacement Resource Recommendation

Purpose

Provide intelligent alternatives when the preferred resource cannot be used.

Trigger Conditions

Recommendations may be generated when:

- Resource unavailable
- Staff absent
- Equipment failed

- Resource under maintenance
- Capacity exhausted
- Certification expired
- Customer-selected resource unavailable
- Venue unavailable

Similarity Matching

For physical resources, TICVAI shall compare:

- Resource type
- Specifications
- Capacity
- Features
- **Location**
- Dependencies

- **Availability**

For staff, TICVAI shall compare:

- Role
- Skill
- Certification
- Language
- Experience
- **Availability**
- Venue

Example

Requested

Projector P-17

Status:

Under Maintenance

Alternatives

P-24 — 98% Match

- ✓ Same specification
- ✓ Same venue
- ✓ Available

P-31 — 92% Match

- ✓ Compatible
- ⚠ Different venue
- ⚠ 30-minute transfer

Customer Impact

Where replacement affects a customer booking, TICVAI shall identify whether:

- No notification required
- Customer notification recommended
- Customer approval required

Acceptance Condition

When a preferred resource is unavailable, TICVAI can automatically identify operationally compatible alternatives and rank them by suitability and impact.

Screen 09 — Operational Scenario Simulator & Digital Twin

Purpose

Allow managers to test operational scenarios before changing the live resource plan.

This should be one of Board 8's signature AI capabilities.

Scenario Creation

Users may create simulations such as:

- **Scenario A**
Attendance increases by 20%.
- **Scenario B**
10 instructors become unavailable.
- **Scenario C**
Main Hall closes unexpectedly.
- **Scenario D**
Add another 2,000-person event on Friday.
- **Scenario E**
Reduce labor cost by 10%.
- **Simulation Engine**
TICVAI shall calculate the impact on:
 - Staffing
 - Equipment
 - Venues
 - Capacity
- **Availability**
 - Conflicts
 - Overtime
 - Costs
 - Utilization
 - Event readiness
 - Customer bookings

Example

Scenario

Attendance +20% Saturday

Impact:

Additional Staff Required: **+14**

Additional Equipment: **+22**

Overtime: **+38 hrs**

Resource Conflicts: 7

Estimated Additional Cost:

AED 11,400

AI Recommendation

Add six temporary staff and redistribute eight existing employees between 12:00 and 18:00.

Projected revised cost:

AED 8,900

Compare Scenarios

Users shall compare:

Current Plan

Scenario A

Scenario B

AI Optimized Scenario

Apply Scenario

An approved simulation may be converted into proposed operational changes.

It must not automatically alter live operations without configured approval.

Acceptance Condition

Managers can safely simulate operational changes and understand resource, workforce, capacity, and financial consequences before modifying live schedules.

Screen 10 — Conversational AI Resource Copilot

Purpose

Allow managers to interact with the complete Resource Management module using natural-language questions and commands.

Example Queries

Users may ask:

“Which resources are underutilized this week?”

“Show me all staffing gaps tomorrow.”

“Who is the best instructor for the 2 PM private lesson?”

“Why is Corporate Gala only 92% resource-ready?”

“Which equipment is likely to cause operational issues this weekend?”

“Can we run another 1,000-person event Friday?”

“How can I reduce overtime tomorrow without affecting operations?”

Context Awareness

The AI Copilot shall understand:

- Current user
- Permission scope
- Selected tenant
- Venue
- Date

- Event
- Resource
- Screen context

Response Types

The Copilot may return:

- Direct answer
- Resource list
- Chart
- Calendar view
- Forecast
- Conflict explanation
- Recommendation
- Scenario
- Suggested action

Actionable AI

Where permitted, users may request:

“Find replacements for all absent staff tomorrow.”

TICVAI shall prepare recommendations.

Before applying changes, the system shall display:

- **Proposed Changes**
- 8 assignments affected
- 7 automatic replacements found
- 1 requires manager action
- **[Review Changes]**
- **[Submit for Approval]**
- **Permission Awareness**

AI shall never allow a user to perform an action outside their normal permissions.

The AI layer must use the same RBAC/PBAC and approval controls as the rest of TICVAI.

Explainability

Users shall be able to ask:

“Why?”

Example:

“Why did you recommend Maria?”

TICVAI:

Maria meets all mandatory qualifications, is already located at the venue, speaks the customer's preferred language, has no conflicting assignments, and does not create overtime.

Acceptance Condition

Authorized users can query, analyze, simulate, and initiate governed resource-management actions conversationally without bypassing normal platform permissions or approval controls.

Board 8 — AI Decision Architecture

TICVAI should not create ten independent AI models for these screens.

Board 8 should operate through a common **Resource Intelligence Layer** consuming trusted operational data from the Resource Management domain.

The conceptual architecture is:

Operational Data

↓

Resource Digital Twin

↓

Availability & Eligibility Engine

↓

Demand Forecasting

↓

Optimization Engine

↓

AI Reasoning & Recommendation Layer

↓

Human Approval / Automation Policy

↓

Operational Execution

This allows the same intelligence to serve the backend, POS, Employee App, Event Management, and other authorized channels.

Board 8 — AI Confidence & Explainability

Every material AI recommendation should expose, where applicable:

- Recommendation
- Confidence
- Factors considered
- Mandatory constraints
- Alternatives
- Operational impact
- Financial impact
- Risks
- Data freshness

Example:

AI Recommendation — 96% Confidence

Transfer 4 instructors from Zone B to Zone A from 14:00–17:00.

Why:

- ✓ Zone A demand forecast +28%
- ✓ Zone B has forecast spare capacity

- ✓ All four instructors are qualified
- ✓ No existing booking conflicts
- ✓ No overtime generated

Expected impact:

Zone A Coverage: 87% → 100%

Zone B Coverage: 118% → 103%

Board 8 — Human-in-the-Loop Governance

AI autonomy shall be configurable by action type.

Recommendation Only

AI proposes an action but cannot execute it.

User Confirmation

AI proposes and an authorized user confirms.

Approval Required

AI proposes, user submits, authorized approver approves.

Auto-Execute

Allowed only for explicitly approved low-risk scenarios.

Examples might include:

- Recommending resources — automatic
- Low-risk resource rebalancing — configurable
- Employee shift changes — approval
- Safety-critical staffing changes — mandatory approval
- Financially significant changes — approval

AI must never bypass:

- Safety rules
- Qualification requirements
- Certification requirements
- Capacity rules
- Legal/workforce rules
- Approval policies
- User permissions

Board 8 — Learning & Feedback

Where approved, TICVAI shall capture outcomes such as:

Recommendation → Accepted / Rejected → Actual Result

This may be used to evaluate and improve future recommendation quality.

Examples:

- Forecast accuracy
- Replacement success
- Staffing forecast accuracy
- Optimization savings achieved

- Conflict-resolution success
- AI performance itself should therefore become measurable.

Board 8 — Shared Governance

All ten screens shall support, where applicable:

- Multi-tenant architecture
- Multi-venue intelligence
- RBAC/PBAC
- Approval workflows
- Immutable audit history
- AI decision logging
- Explainability
- Confidence indicators
- Data lineage
- Model/version traceability where applicable
- Arabic and English
- API accessibility
- Event-driven architecture
- Human override
- Sensitive-data controls

Every executed AI-generated action must be distinguishable in the audit history from a manually initiated action.

Board 8 — End-to-End Outcome

After Board 8, TICVAI Resource Management changes from a system that **records operational information** into a system that actively helps operate the business.

Instead of the manager discovering tomorrow morning:

“We don’t have enough instructors.”

TICVAI should identify the issue beforehand:

Tomorrow's Resource Forecast

Expected Attendance

8,420

Resource Readiness

94%

Predicted Issues

△ 5 staffing gaps

△ 2 equipment shortages

△ 18 overtime hours

TICVAI Optimization Available

✓ Qualified replacements identified for 4 staffing gaps

- ✓ Equipment can be transferred from Venue B
- ✓ Overtime can be reduced by 11 hours

Estimated saving: AED 3,450

Recommended Action

Review AI Optimized Plan

The full Board 8 intelligence journey therefore becomes:

Observe → Predict → Detect → Recommend → Simulate → Optimize → Explain → Approve → Execute → Learn

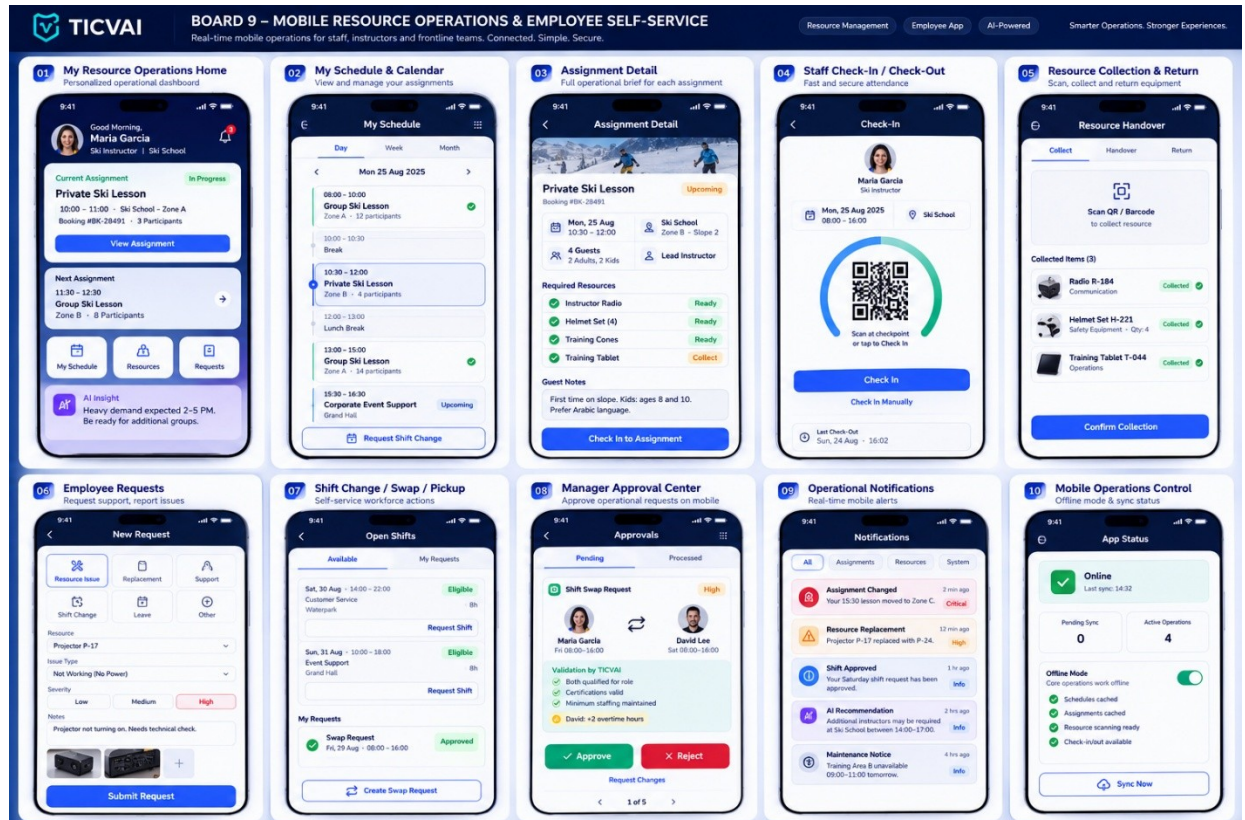
That is what should make TICVAI Resource Management feel fundamentally different from a conventional scheduling or resource-booking platform.

TICVAI Resource Management

Board 9 — Mobile Resource Operations & Employee Self-Service

Board Objective

Provide TICVAI employees and operational managers with a mobile-first **Resource Operations Workspace** where they can see their schedules and assignments, receive operational instructions, check in and out, manage resource handovers, submit workforce requests, approve operational actions, and receive real-time notifications.



Board 9 shall expose the appropriate functions through the **TICVAI Employee App** rather than creating a separate Resource Management mobile application.

The mobile experience shall consume the same centralized services used by the TICVAI backend:

Resource Master → Calendar → Workforce → Assignment → Attendance → Assets → Events → AI → Mobile

Therefore, an assignment changed in the backend must immediately appear in the employee's application, subject to synchronization and connectivity.

The experience must be designed for frontline operations:

- Minimal typing
- Large touch targets
- Scan-first workflows
- Clear status
- Fast confirmation
- Mobile notifications
- Offline support where required

- Role-specific interfaces
- Real-time synchronization
- Arabic and English

Primary matrix coverage: 1.2.60–1.2.66.

Supporting coverage: 1.2.11, 1.2.17, 1.2.30–1.2.38, 1.2.52, 1.2.60–1.2.66, 1.2.80–1.2.81.

Screen 01 — My Resource Operations Home

Purpose

Provide each employee with a personalized operational landing screen showing what requires their attention now and next.

Home Dashboard

The screen shall display:

- Current shift
- Current assignment
- Next assignment
- Today's schedule
- Assigned venue
- Check-in status
- Break status
- Pending tasks
- Resource handovers
- Employee requests
- Notifications
- Manager messages
- **Smart Operational Card**
The primary card should always answer:
“What am I supposed to do now?”

Example:

NOW

Private Ski Lesson

14:00–15:00

Ski School — Zone A

Customer: Booking #BK-28491

Participants: 3

Status:

Starts in 12 min

Actions:

[View Assignment]

[Check In]

Next Assignment

NEXT

Group Ski Lesson

15:30–16:30

Zone B

Participants: 8

Alerts

Examples:

⚠ Your 15:30 assignment has moved to Zone C.

✓ Equipment package is ready for collection.

⚠ Your certification expires in 14 days.

AI Assistant

The employee may ask:

“What is my next assignment?”

or:

“Where do I need to go after this?”

Acceptance Condition

Employees can immediately understand their current operational status and next required action without navigating through multiple screens.

Screen 02 — My Schedule & Assignment Calendar

Purpose

Provide employees with a mobile view of their working schedule and operational assignments.

Calendar Views

Support:

- Today
- Tomorrow
- Week
- Agenda
 - **Schedule Timeline**

Example:

 - **Tuesday — 25 August**
 - **08:00–10:00**
 - Group Ski Lesson
 - Zone A
 - **10:00–10:30**
 - Break
 - **10:30–12:00**
 - Private Lesson
 - Zone B
 - **12:00–13:00**
 - Lunch
 - **13:00–15:00**
 - Available

- **15:00–17:00**

- Corporate Event Support
- Grand Hall

- **Assignment Status**

Display:

- Upcoming
- Confirmed
- In Progress
- Completed
- Changed
- Cancelled
- Requires Attention

- **Live Updates**

When a manager changes an assignment:

Old Assignment

Zone A

→

New Assignment

Zone C

The employee shall receive an immediate update and notification.

Acceptance Condition

Employees can view their complete operational schedule and receive synchronized updates whenever assignments change.

Screen 03 — Assignment Detail & Operational Brief

Purpose

Give the employee all information required to execute a specific assignment.

Assignment Header

Display:

- Assignment
- Date
- Start/end
- Venue
- Zone/location
- Role
- Status
- Priority

- **Operational Information**

Depending on assignment type:

- Event
- Experience
- Customer booking
- Participant count

- Required resource
- Equipment
- Instructions
- Checklist
- Contact person
- Supervisor

Location

Where appropriate, provide:

- Venue map
- Zone
- Meeting point
- Navigation link within configured venue mapping

Required Resources

Example:

Equipment Required

- ✓ Instructor Radio
- ✓ Helmet Set
- ✓ Training Cones

⚠ Training Tablet — Collection Required

Customer Information

Only operationally necessary customer information shall be shown.

Sensitive customer data must remain permission controlled.

Quick Actions

- Start assignment
- Check in
- Contact supervisor
- Report issue
- Request replacement
- View checklist
- Complete assignment

Acceptance Condition

Employees can access all permitted operational information needed to execute an assignment from one mobile screen.

Screen 04 — Mobile Staff Check-In & Check-Out

Purpose

Allow employees to record attendance and assignment presence from the Employee App.

Check-In

The system shall support configured verification methods such as:

- Mobile confirmation

- QR
- NFC
- Venue QR
- Resource QR
- Manager validation
- Integrated attendance system

- **Check-In Screen**

Display:

- **Shift**
- 08:00–16:00
- **Venue**
- Ski School
- **Current Time**
- 07:56
- **Status**
- **Ready to Check In**
- **[CHECK IN]**

Validation

Where configured, TICVAI may validate:

- Correct date/time
- Venue
- Scheduled shift
- Assignment
- Device/session
- Allowed check-in window

Assignment Check-In

Separate assignment-level check-in may be supported.

Example:

Check into Private Ski Lesson — Zone A

Check-Out

On checkout, display:

- Actual working time
- Breaks
- Overtime
- Exceptions
- Outstanding assignments

- **Offline Mode**

Where required, check-in/out may be captured offline with:

- Device timestamp
- Offline indicator
- Later synchronization

Conflict-resolution rules shall apply when synchronization occurs.

Acceptance Condition

Employees can record shift and assignment attendance from the mobile application with configurable validation and full audit history.

Screen 05 — Resource Collection, Handover & Return

Purpose

Digitally manage physical resources transferred between employees, departments, or operational locations.

Example

An event technician receives:

- Radio R-184
- Tablet T-044
- AV Toolkit K-12

Collection

Employee scans:

Resource QR

TICVAI displays:

Radio R-184

Status: Available

Assigned to:

Corporate Gala

Employee:

Ahmed Khan

Actions:

[Accept Resource]

Handover

For employee-to-employee transfer:

Ahmed

→ Radio R-184 →

Sarah

Both parties may be required to confirm.

Condition

Capture:

- Good
- Minor damage
- Damaged
- Missing component

Photo evidence may be added where configured.

- **Return**

On return:

- Scan resource

- Confirm condition
- Confirm location
- Submit

Resource status shall update immediately.

- **Chain of Custody**

The resource timeline shall record:

Storage → Employee A → Employee B → Event → Returned

Acceptance Condition

Physical-resource custody can be digitally tracked through scan-driven collection, handover, and return workflows.

Screen 06 — Employee Requests & Resource Support

Purpose

Allow frontline employees to submit operational resource-related requests directly from the Employee App.

Request Types

Support configurable requests such as:

- Replacement resource
- Additional equipment
- Resource issue
- Maintenance request
- Assignment change
- Assistance required
- Venue change request
- Schedule clarification
- Custom operational request

Example

Resource Issue

Resource:

Projector P-17

Issue:

No Power

Severity:

High

Photo:

Attached

Action:

[Submit Issue]

Immediate Operational Impact

If the issue makes the resource unusable, authorized workflows may:

1. Mark resource unavailable.

2. Create maintenance request.
3. Identify affected assignments.
4. Trigger AI replacement search.
5. Alert manager.

- **Request Tracking**

Employees shall see:

- Submitted
- Under Review
- Approved
- Rejected
- Resolved

Acceptance Condition

Employees can report operational resource issues and requests digitally, and those requests can trigger the appropriate backend workflow.

Screen 07 — Shift Change, Swap, Pickup & Release

Purpose

Expose the Board 4 Shift Marketplace capabilities through the Employee App.

This screen does not recreate shift rules; it consumes the workforce rules already configured in Board 4.

My Shift Actions

Employees may request:

- Swap
- Transfer
- Release

Open Shifts

Eligible employees may view available shifts.

Example:

Open Shift

Saturday

14:00–22:00

Waterpark

Role:

Customer Service

You are eligible

Estimated hours:

8h

[Request Shift]

Eligibility

Before displaying or allowing a shift request, TICVAI shall evaluate:

- Role

- Skill
- Certification
- **Availability**
- Working hours
- Rest
- Overtime
- Venue permission

Shift Swap

Employee selects:

My Shift

Friday 08:00–16:00

and:

Requested Employee / Open Exchange

TICVAI validates both sides before submission.

Approval

Display request status:

Awaiting Employee

→ **Awaiting Manager**

→ **Approved**

Acceptance Condition

Employees can manage permitted shift changes through mobile self-service while all qualification, compliance, and approval rules remain enforced centrally.

Screen 08 — Manager Mobile Approval Center

Purpose

Allow authorized supervisors and managers to approve operational requests without requiring access to the desktop backend.

Approval Inbox

Display requests such as:

- Shift swap
- Shift pickup
- Overtime
- Resource replacement
- Assignment change
- Resource transfer
- Maintenance block
- Attendance correction
- Operational exception
 - **Approval Card**
Example:
 - **Shift Swap Request**

- Maria Garcia
- Friday 08:00–16:00
- ↔
- David Lee
- Saturday 08:00–16:00

TICVAI Validation

- ✓ Both qualified
- ✓ Certifications valid
- ✓ Minimum staffing maintained
- ✓ Rest rules satisfied
- ⚠ David: +2 overtime hours

Actions**[Approve]****[Reject]****[Request Changes]****AI Summary**

For complex requests:

TICVAI Recommendation: Approve

All mandatory rules pass. The change creates two overtime hours but maintains full operational coverage.

Governance

Approval authority shall use the same RBAC/PBAC rules configured in the backend.

Acceptance Condition

Authorized managers can review and action operational approvals from mobile with sufficient validation context to make an informed decision.

Screen 09 — Operational Notifications & Live Alerts**Purpose**

Provide employees and managers with timely, prioritized operational notifications.

Notification Categories

Support:

- Assignment
- Schedule
- Shift
- Resource
- Maintenance
- Event
- Attendance
- Approval
- Emergency/critical operations
- AI recommendation

- **Priority**
Notifications may be:
- **Critical**
Immediate operational action.
- **High**
Requires timely response.
- **Standard**
Operational information.
- **Informational**
No action required.

Examples

Critical

Assignment Changed

Corporate Gala AV assignment moved to Main Hall. Start time unchanged.

High

Resource Replacement

Projector P-17 is unavailable. P-24 has been assigned.

Standard

Shift Approved

Your Saturday shift pickup has been approved.

Notification Actions

Users may:

- Open
- Acknowledge
- Accept
- Decline where permitted
- Navigate to affected assignment
 - **Escalation**
Critical alerts may require acknowledgement.
If not acknowledged within a configured period, TICVAI may escalate to:
 - Supervisor
 - Operations manager
 - Other configured recipient

Acceptance Condition

Operational changes and critical resource events are communicated to the correct users with configurable priority, acknowledgement, and escalation behavior.

Screen 10 — Mobile Operations Control & Offline Sync

Purpose

Configure and monitor how Resource Management operates on mobile devices, particularly where connectivity is unreliable.

Mobile Configuration

Administrators shall control:

- Offline capability
- Offline assignment cache
- Offline schedule cache
- Check-in offline
- Resource scanning offline
- Handover offline
- Checklist offline
- Maximum offline period
- Synchronization frequency
- Data-retention policy

Offline Workspace

When disconnected:

OFFLINE MODE

Last synchronized:

14:32

Cached:

- ✓ Today's assignments
- ✓ Resource details
- ✓ Operational checklist
- ✓ Venue information

Pending synchronization:

4 actions

Pending Actions

Example:

1. Staff check-in
2. Resource handover
3. Assignment completion
4. Resource issue report

Reconnection

On reconnection:

Synchronizing 4 actions...

TICVAI shall validate each transaction against current server state.

Conflict Resolution

Example:

A resource was assigned offline but became unavailable centrally.

TICVAI shall not silently overwrite the conflict.

It shall:

1. Flag synchronization conflict.
2. Preserve original offline transaction.

3. Determine operational impact.
4. Request resolution or apply configured conflict policy.

Device & Sync Monitoring

Administrators shall be able to see:

- Active mobile sessions
- Last sync
- Offline devices
- Pending transactions
- Sync failures
- Version status

Acceptance Condition

The Employee App can support configured resource operations during temporary connectivity loss and safely synchronize transactions when connectivity returns without compromising the authoritative resource state.

Board 9 — Role-Based Mobile Experience

Board 9 should not show every employee the same application.

The mobile interface shall adapt based on role.

Instructor

My Schedule → Lessons → Check-In → Equipment → Requests

Event Technician

Event Assignments → Equipment Collection → Checklist → Handover → Issues

Frontline Rental Employee

Resource Checkout → Return → Inspection → Issue Reporting

Supervisor

Team Status → Staffing Gaps → Approvals → Alerts → Reassignments

Operations Manager

Resource Readiness → Critical Alerts → Approvals → AI Recommendations

This should be driven by permission and operational context rather than separate applications.

Board 9 — Smart Mobile Home Principle

The Employee App should prioritize **contextual actions**, not menus.

If the employee has an assignment starting in ten minutes, the application should immediately surface:

NEXT ACTION

Private Ski Lesson — 14:00

Zone A

Equipment Ready ✓

Customer Checked In ✓

[Start Assignment]

If instead the employee has an unreturned resource:

ACTION REQUIRED

Radio R-184

Return due at 18:00

[Return Resource]

The employee should not have to search the application to determine what needs attention.

Board 9 — Mobile AI Copilot

The Resource AI layer from Board 8 should be available in a controlled form through the Employee App.

Employees may ask:

“Where is my next assignment?”

“Which equipment do I need?”

“Can I swap my Friday shift?”

“Who should I hand this radio to?”

Managers may ask:

“Who hasn't checked in?”

“Show me today's staffing gaps.”

“Find a replacement for Ahmed.”

The Copilot shall only return information and actions the logged-in user is permitted to access.

Board 9 — Offline-First Architecture

Because TICVAI may operate in attractions, stadiums, theme parks, event venues, and temporary event environments, selected Employee App capabilities should be designed to tolerate unstable connectivity.

The architecture should therefore support, where required:

Server State

↓

Encrypted Local Operational Cache

↓

Offline Action Queue

↓

Connectivity Restored

↓

Validation & Conflict Detection

↓

Server Synchronization

Offline support must never create uncontrolled duplicate assignments or conflicting resource states.

Board 9 — Shared Security & Governance

All mobile screens shall support, where applicable:

- RBAC/PBAC
- Tenant isolation
- Venue restrictions
- Device/session security
- SSO
- MFA where required
- Encrypted local storage
- API security
- Customer-data minimization
- Immutable audit history
- Mobile action traceability
- Offline transaction traceability
- Arabic and English
- Accessibility
- Approval workflows
- Push notifications

Sensitive information must be limited to what the employee requires to perform the assignment.

Board 9 — Cross-Platform Synchronization

Board 9 shall use the same underlying Resource Management services as:

- Backend Administration
- POS
- B2C
- B2B
- Event Management
- Workforce Management
- Equipment & Asset Management
- AI Resource Intelligence
- APIs

For example:

Manager changes Maria's assignment in Backend

- Resource Calendar updates
- Workforce Roster updates
- Maria's Employee App updates
- Push notification generated
- Old resource availability released
- New resource availability reserved
- Audit event recorded

No separate mobile resource database shall become an independent source of truth.

Board 9 — End-to-End Outcome

After Board 9, TICVAI Resource Management no longer stops at the manager's desktop.

The operational loop becomes:

Manager Plans

- TICVAI Assigns
- Employee Receives
- Employee Checks In
- Employee Executes
- Resource Is Collected / Handed Over
- Employee Reports Exceptions
- Manager Approves / AI Recommends
- Assignment Updates
- Employee Completes
- Backend Updates in Real Time

The employee experience should ultimately be extremely simple:

GOOD MORNING, MARIA

Your Shift

08:00–16:00

Next Assignment

Private Ski Lesson

14:00 — Zone A

Starts in 12 minutes

- ✓ Customer checked in
- ✓ Training area ready
- ✓ Equipment ready

[START ASSIGNMENT]

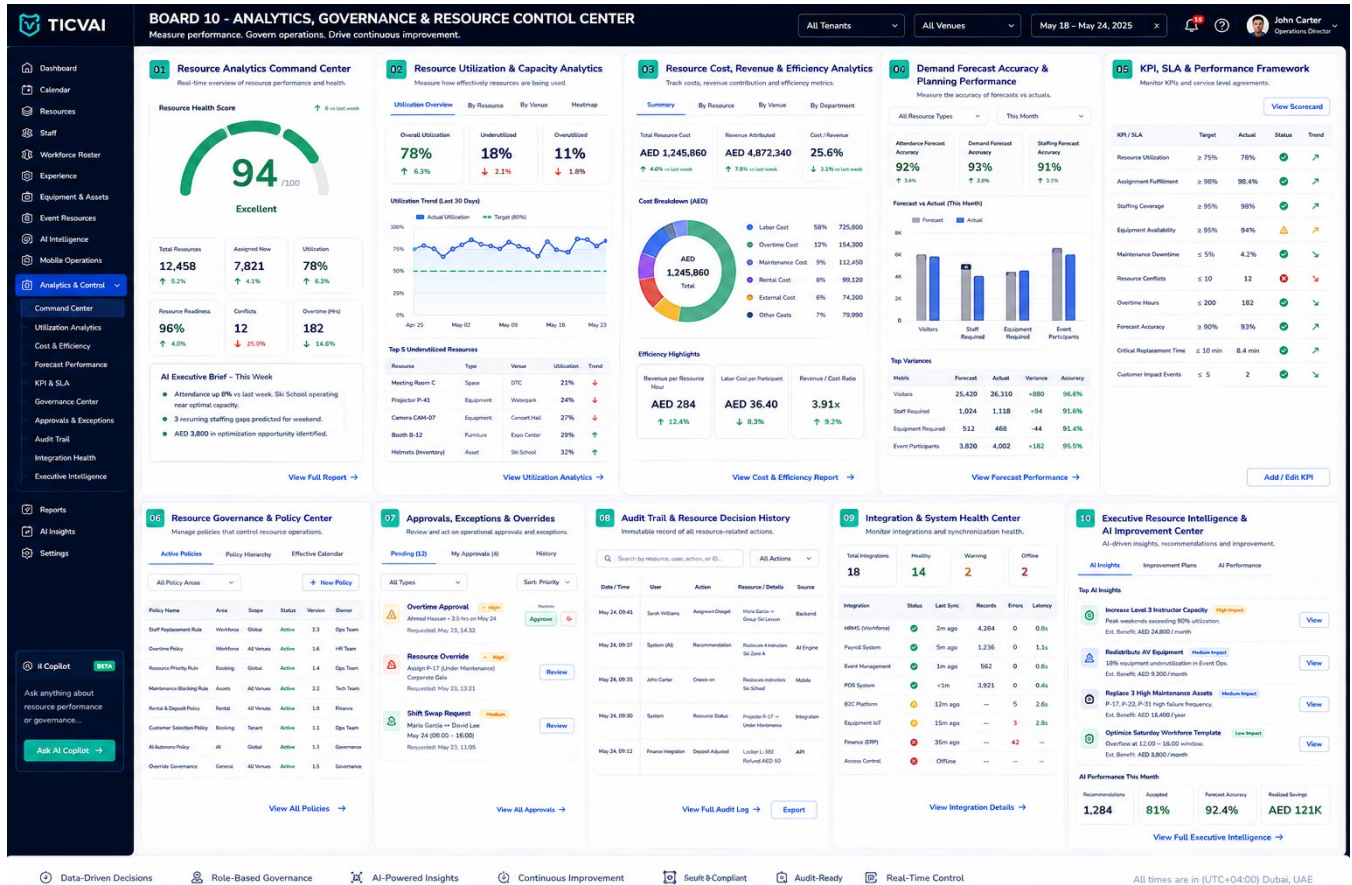
Everything complex—availability, certification, resource dependency, workforce compliance, customer booking, equipment status, and scheduling—has already been validated behind that single button.

That is the objective of Board 9: **enterprise-grade Resource Management intelligence delivered to frontline employees through an extremely simple digital experience.**

Board 10 — Analytics, Governance & Resource Control Center

Board Objective

Provide TICVAI with the executive and administrative control layer for the entire Resource Management ecosystem.



Board 10 shall consolidate information from all previous boards to answer four fundamental questions:

1. Are our resources being used efficiently?
2. Are they generating the expected operational and financial value?
3. Are resource decisions compliant, controlled, and auditable?
4. What should management change to improve future performance?

Board 10 shall therefore cover:

Analytics → Utilization → Cost & Performance → Forecast Accuracy → SLA/KPI → Governance → Approvals → Audit → Integration Health → Executive Intelligence

The board should not recreate operational functions from Boards 1–9. It should provide the **management, governance, analytics, and control plane** above them.

Screen 01 — Resource Analytics Command Center

Purpose

Provide executives and operational managers with a single enterprise dashboard showing the overall health and performance of Resource Management.

Executive KPIs

Display:

- Total active resources
- Available resources
- Currently assigned
- Utilization %
- Resource readiness %
- Staff utilization
- Physical asset utilization
- Resource conflicts
- Unfulfilled resource demand
- Overtime
- Maintenance downtime
- Resource-related operational cost
- Forecast vs actual demand
- AI recommendations implemented
 - **Resource Breakdown**

Allow analysis by:

- Tenant
- Region
- Venue
- Attraction
- Event
- Department
- Resource category
- Resource type
- Staff
- Equipment
- Space
- Date range
- **Executive Health Indicators**

Example:

- **Resource Readiness — 96%**
- **Workforce Coverage — 98%**
- **Asset Availability — 94%**
- **Conflict Resolution — 91%**
- **Forecast Accuracy — 93%**
- **AI Executive Summary**

TICVAI shall generate a concise operational brief such as:

Resource utilization increased 7% this week. Ski School is operating near optimal capacity, while Event Operations has 18% underutilized equipment. Three recurring staffing gaps are predicted for the coming weekend.

Acceptance Condition

Authorized management users can understand overall Resource Management performance and identify areas requiring attention from a single dashboard.

Screen 02 — Resource Utilization & Capacity Analytics

Purpose

Measure how effectively resources are being utilized and identify overused or underused capacity.

Utilization Metrics

For each resource calculate, where applicable:

Available Time

Reserved Time

Assigned Time

Actual Usage

Idle Time

Blocked Time

Maintenance Time

Utilization Formula

For example:

Actual Utilized Hours / Operationally Available Hours

Visualization

Display utilization through:

- Heatmaps
- Trend charts
- Resource rankings
- Venue comparison
- Day/time analysis
- Capacity curves

Example

Ski Instructors — Saturday

Available Capacity: 420 hrs

Scheduled: 376 hrs

Actual: 354 hrs

Utilization:

84.3%

Underutilization

TICVAI shall identify resources consistently below configured utilization targets.

Example:

Meeting Room C averages 21% utilization over the last 90 days.

Overutilization

Example:

Level 3 instructors have operated above 92% utilization for five consecutive weekends.

This may indicate capacity risk rather than good performance.

AI Recommendation

Consider increasing Level 3 instructor capacity for weekend operations.

Acceptance Condition

Managers can understand resource utilization, idle capacity, overutilization, and capacity pressure across the organization.

Screen 03 — Resource Cost, Revenue & Efficiency Analytics

Purpose

Measure the financial and operational efficiency of resources.

Cost Metrics

Support analysis of:

- Labor cost
- Overtime cost
- Maintenance cost
- Rental cost
- External-resource cost
- Transfer cost
- Asset operating cost
- Resource replacement cost

Revenue Association

Where commercial relationships exist, TICVAI shall associate resources with:

- Ticket revenue
- Experience revenue
- Rental revenue
- Event revenue
- Upsell revenue

The Finance module remains the financial system of record where applicable.

Efficiency Metrics

Examples:

Revenue per Resource Hour

Cost per Participant

Labor Cost per Experience

Maintenance Cost per Asset

Revenue / Resource Cost

Example

Private Ski Lessons

Revenue: AED 420,000

Instructor Cost: AED 112,000

Resource Hours: 1,480

Revenue / Instructor Hour:

AED 284

AI Insight

Private lessons generate 34% more revenue per instructor hour than group lessons during weekday afternoons.

The AI should present this as management insight rather than automatically changing commercial policies.

Acceptance Condition

Authorized users can analyze resource-related costs and operational/commercial efficiency while maintaining clear ownership boundaries with Finance.

Screen 04 — Demand Forecast Accuracy & Planning Performance

Purpose

Measure how accurately TICVAI's forecasting and planning engines predicted actual resource demand.

Comparison

Display:

- **Forecast Demand**

vs.

- **Planned Resources**

vs.

- **Actual Demand**

vs.

- **Actual Resources Used**

Metrics

Track:

- Attendance forecast accuracy
- Resource demand forecast accuracy
- Staffing forecast accuracy
- Equipment forecast accuracy
- Forecast shortage rate
- Overstaffing rate
- Understaffing rate

Example

Saturday — Waterpark

Forecast Visitors: 5,200

Actual Visitors: 5,410

Forecast Accuracy:

96.1%

Forecast Lifeguards Required: 21

Actual Requirement: 22

Trend Analysis

Managers shall see whether forecast quality improves over time.

AI Learning Feedback

Where permitted:

Forecast → Actual → Variance → Learning Signal

shall feed model evaluation and future forecasting improvement.

Acceptance Condition

TICVAI can measure forecast accuracy and provide management with evidence of whether AI-assisted resource planning is improving operational performance.

Screen 05 — Resource KPI, SLA & Performance Framework

Purpose

Allow organizations to define measurable Resource Management performance standards.

KPI Library

Administrators may configure KPIs such as:

- Resource utilization
- Assignment fulfillment
- Staffing coverage
- Resource readiness
- Equipment availability
- Maintenance downtime
- Conflict resolution time
- Resource replacement time
- Attendance compliance
- Overtime
- Rental return rate
- Forecast accuracy

KPI Configuration

Each KPI shall support:

- Name
- Description
- Formula
- Target
- Warning threshold
- Critical threshold
- Applicable venue
- Applicable resource
- Effective period
- Owner

Example

Resource Assignment Fulfillment

Target:

≥ 98%

Warning:

95–97.9%

Critical:

<95%

SLA Monitoring

Example:

Critical Resource Replacement

Target:

Replacement identified within **10 minutes**

Scorecards

Display:

- **Target | Actual | Variance | Trend | Status**

Acceptance Condition

Administrators can configure Resource Management KPIs and SLAs and management can continuously monitor performance against those targets.

Screen 06 — Resource Governance & Policy Center

Purpose

Centralize the administrative policies controlling how Resource Management behaves.

Policy Areas

Administrators shall manage policies relating to:

- Resource assignment
 - **Availability**
 - Booking
 - Resource priority
 - Customer selection
 - Qualification
 - Certification
 - Overtime
 - Maintenance blocking
 - Rental
 - Deposits
 - Resource replacement
 - Event allocation
 - AI autonomy
 - Overrides

Policy Structure

Each policy shall support:

- Policy name
- Scope
- Tenant
- Venue
- Resource category
- Effective date
- Expiry date
- Owner
- Version
- Approval requirement

Policy Hierarchy

TICVAI shall support inheritance such as:

- **Global**
- ↓
- **Tenant**
- ↓
- **Region**
- ↓
- **Venue**
- ↓
- **Resource Category**
- ↓
- **Specific Resource**

The interface must clearly show inherited versus locally overridden values.

Example

Automatic Staff Replacement

Global: Allowed

Venue A: Allowed

Venue B: **Manager Approval Required**

Acceptance Condition

Administrators can centrally manage Resource Management policies with controlled inheritance, versioning, and effective dating.

Screen 07 — Approval, Exception & Override Control Center

Purpose

Provide one centralized workspace for governed Resource Management exceptions.

Approval Types

Include:

- Resource override

- Qualification exception
- Overtime
- Resource conflict
- Shift change
- Event allocation
- Maintenance override
- Rental adjustment
- Deposit adjustment
- AI-generated operational change

Approval Queue

Each request shall show:

- Request
- Resource
- Requester
- Venue
- Reason
- Operational impact
- Financial impact
- Risk
- AI recommendation
- Required approver

Example

Resource Override

Request

Assign David to Corporate Event A.

Issue

Would create 2 overtime hours.

Mandatory Rules

- ✓ Qualification
- ✓ Certification
- ✓ Rest period

AI Recommendation

Approve with Overtime

Actions

Approve

Reject

Request Changes

Escalate

Non-Overrideable Rules

Safety-critical or legally mandatory policies may be configured so that no user can override them.

Acceptance Condition

Resource exceptions and overrides are processed through controlled, permission-based workflows rather than informal administrative changes.

Screen 08 — Audit Trail & Resource Decision History

Purpose

Provide immutable traceability of significant Resource Management changes and decisions.

Audit Events

Capture:

- Resource creation
- Resource modification
- Availability change
- Assignment
- Reassignment
- Cancellation
- Staff override
- Maintenance block
- Rental transaction
- Deposit adjustment
- Event allocation
- AI recommendation
- AI execution
- Approval
- Policy change

Audit Record

Each event shall capture:

- **Who**
- **What**
- **When**
- **Where**
- **Previous Value**
- **New Value**
- **Reason**
- **Approval**
- **Source**

Source Identification

Actions shall clearly indicate whether initiated by:

- User
- Employee App
- POS
- API

- **Integration**
- Automated rule
- TICVAI AI

Example

Assignment Changed

Resource: Maria Garcia

Original: Private Lesson — Zone A

New: Group Lesson — Zone B

Initiated By: TICVAI AI Recommendation

Approved By: Operations Manager

Reason: Critical staffing gap

Search

Users shall search audit history by:

- Resource
- Employee
- Event
- User
- Date
- Action
- Venue
- Source

Acceptance Condition

Authorized users can reconstruct the complete history of material Resource Management decisions and identify exactly how and why each change occurred.

Screen 09 — Resource Integration & System Health Center

Purpose

Monitor all technical integrations and synchronization services supporting Resource Management.

Connected Systems

Potential connections include:

- HRMS
- Workforce Management
- Payroll
- Event Management
- Ticketing
- POS
- B2C
- Employee App
- Finance
- Inventory
- Procurement

- Access Control
- External APIs

Integration Dashboard

Display:

- **System**
- **Status**
- **Last Sync**
- **Transactions**
- **Errors**
- **Latency**

Health Status

Use clear indicators:

Healthy

Warning

Degraded

Offline

Example

HRMS

Status: **Healthy**

Last Sync: 2 min ago

Records: 4,284

Errors: 0

Workforce Provider

Status: **Warning**

Last Sync: 18 min ago

Failed Records: 12

Operational Impact

This is important.

TICVAI shall not simply say:

“Integration failed.”

It should explain:

12 employee availability updates could not be synchronized. Four employees have assignments within the next 24 hours.

Actions

Authorized users may:

- Retry
- View error
- Inspect payload/reference
- Reprocess
- Escalate
- Open affected records

Acceptance Condition

Administrators can monitor integration health and understand the operational impact of synchronization failures.

Screen 10 — Executive Resource Intelligence & AI Improvement Center

Purpose

Complete the Resource Management module with an executive AI workspace that converts operational data into management recommendations.

This should be the **hero screen of Board 10**.

Executive Question

Management should be able to ask:

“How did our resources perform this month?”

TICVAI may respond:

Resource Performance — August

Overall Utilization

78% ↑ 6%

Resource Readiness

96%

Assignment Fulfillment

98.4%

Forecast Accuracy

92%

Overtime

↓ 14%

Resource-Related Cost

↓ 4.8%

AI Findings

1. Ski School Capacity Pressure

Level 3 instructors exceeded 90% utilization on four weekends.

2. Event Equipment Underutilization

AV equipment utilization averaged 38%.

3. Maintenance Risk

Three projectors account for 41% of AV maintenance expenditure.

4. Workforce Optimization

AI roster recommendations reduced estimated overtime by 64 hours.

Recommended Actions

TICVAI may recommend:

Increase Level 3 instructor capacity

Expected effect:

+8% peak availability

Redistribute AV equipment

Expected effect:

+12% utilization

Replace Projectors P-17, P-22 and P-31

Expected effect:

AED 18,400 estimated annual maintenance reduction

Adjust Saturday workforce templates

Expected effect:

-11 overtime hours/week

Action Governance

Each recommendation shall show:

- Evidence
- Confidence
- Expected benefit
- Risk
- Affected resources
- Financial impact
- Operational impact

Management may:

- **Review**
- **Simulate**
- **Create Action**
- **Assign Owner**
- **Dismiss**

The AI should never silently make strategic management changes.

Acceptance Condition

Executives can use Resource Management data and AI intelligence to identify measurable opportunities for improving capacity, cost, reliability, utilization, and operational performance.

Board 10 — Resource Management Control Tower

Board 10 effectively becomes the **Control Tower** sitting above the previous nine boards.

The architecture should conceptually operate as:

Board 1-2

Resource Master + Availability

↓

Board 3-4

Workforce + Rostering

↓

Board 5

Experience & Dynamic Assignment

↓

Board 6

Equipment & Physical Assets

↓

Board 7

Event Resource Planning

↓

Board 8

AI Forecasting & Optimization

↓

Board 9

Mobile Operations

↓

BOARD 10

ANALYZE → GOVERN → AUDIT → IMPROVE

Board 10 should therefore consume information from all Resource Management services rather than creating another independent data store.

Board 10 — Executive Resource Health Score

TICVAI should calculate a configurable overall **Resource Health Score**.

For example:

RESOURCE HEALTH

94 / 100

Availability — 97%

Utilization — 88%

Workforce Coverage — 98%

Asset Reliability — 91%

Event Readiness — 96%

Forecast Accuracy — 93%

Compliance — 100%

The exact weighting shall be configurable by organization.

Selecting any component should drill down into the underlying issues.

Board 10 — AI Performance Governance

Because AI plays an important role throughout Resource Management, TICVAI should also measure the AI itself.

Track:

- Recommendations generated

- Recommendations accepted
- Recommendations rejected
- Auto-actions executed
- Forecast accuracy
- Optimization savings predicted
- Optimization savings realized
- Replacement success
- Conflict-resolution success

Example:

- **AI Resource Intelligence — August**
- Recommendations: **1,284**
- Accepted: **81%**
- Rejected: **12%**
- Expired/No Action: **7%**
- Forecast Accuracy: **92.4%**
- Estimated Savings: **AED 148,000**
- Validated Savings: **AED 121,000**

This gives management measurable evidence of the value delivered by TICVAI's AI layer.

Board 10 — Governance Architecture

All configuration and operational actions across Resource Management shall remain governed by:

Identity

→ RBAC/PBAC

→ Policy

→ Validation

→ Approval

→ Execution

→ Audit

→ Analytics

AI shall operate inside this governance chain rather than outside it.

Board 10 — Shared Security & Administration

All screens shall support, where applicable:

- Multi-tenant architecture
- Multi-region operation
- Multi-venue operation
- RBAC/PBAC
- Maker-checker workflows
- Effective dating
- Versioning
- Immutable audit history

- Data retention policies
- API security
- Integration monitoring
- AI explainability
- AI decision logging
- Data lineage
- Arabic and English
- Accessibility
- Export permissions
- Sensitive-data controls

Board 10 — End-to-End Outcome

After Board 10, the entire Resource Management lifecycle is complete.

TICVAI can now move from:

Create Resource

- Define Availability
- Configure Workforce
- Build Roster
- Assign to Experience
- Manage Physical Asset
- Allocate to Event
- Predict & Optimize
- Operate through Mobile
- Measure & Govern

And management should ultimately receive something as simple as:

RESOURCE HEALTH — 94/100

Tomorrow

96% Resource Ready
 3 predicted shortages
 2 maintenance risks
 11 overtime hours avoidable
 AED 3,800 optimization opportunity

TICVAI Recommendation

5 actions identified

4 can be resolved using existing resources

1 requires management decision

[Review AI Action Plan]